

VisuLogic: Benchmark for Evaluating Visual Reasoning in Multi-modal Large Language Models

Weiye Xu^{3†}, Jiahao Wang^{2 3†}, Weiyun Wang^{3†}, Zhe Chen^{3†}, Wengang Zhou^{*}, Jijun Yang²,
Lewei Lu⁴, Houqiang Li^{*}, Xiaohua Wang², Xizhou Zhu³, Wenhai Wang³, Jifeng Dai^{5 3}, Jinguo Zhu³

University of Science and Technology of China, ²Xi'an Jiaotong University,
³Shanghai Artificial Intelligence Laboratory, ⁴SenseTime Research, ⁵Tsinghua University
ustcxwy0271@mail.ustc.edu.cn, wjhwscience@stu.xjtu.edu.cn, lechatelia@gmail.com

Abstract

Visual reasoning is a core component of human intelligence and a critical capability for advanced multimodal models. Yet current reasoning evaluations of multimodal large language models (MLLMs) often rely on text descriptions and allow language-based reasoning shortcuts, failing to measure genuine vision-centric reasoning. To address this, we introduce VisuLogic: a benchmark of 1,000 human-verified problems across six categories (e.g., quantitative shifts, spatial relations, attribute comparisons). These various types of questions can be evaluated to assess the visual reasoning capabilities of MLLMs from multiple perspectives. We evaluate leading MLLMs on this benchmark and analyze their results to identify common failure modes. Most models score below 30% accuracy—only slightly above the 25% random baseline and far below the 51.4% achieved by humans—revealing significant gaps in visual reasoning. Furthermore, we provide a supplementary training dataset and a reinforcement-learning baseline to support further progress. Code, data, and baselines are available at <https://visulogic-benchmark.github.io/VisuLogic>.

1 Introduction

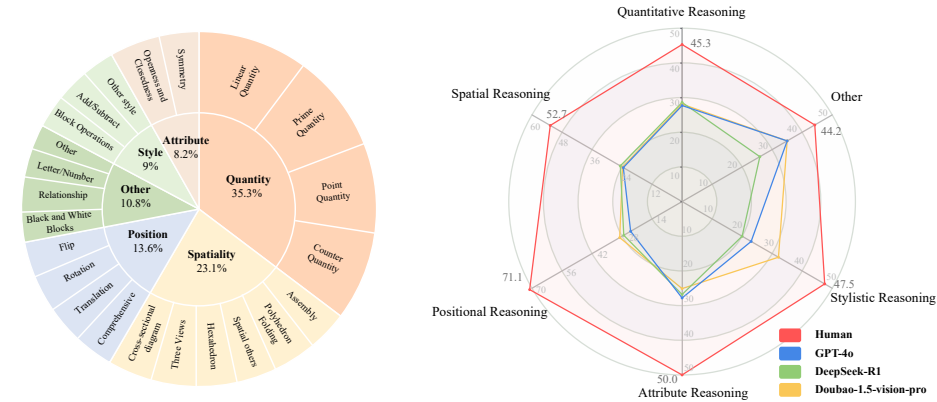
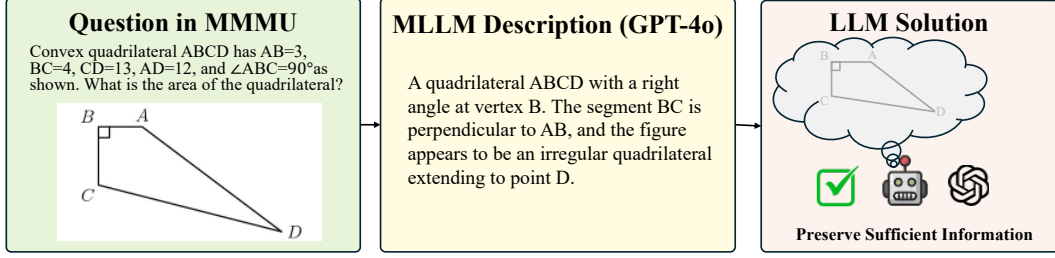
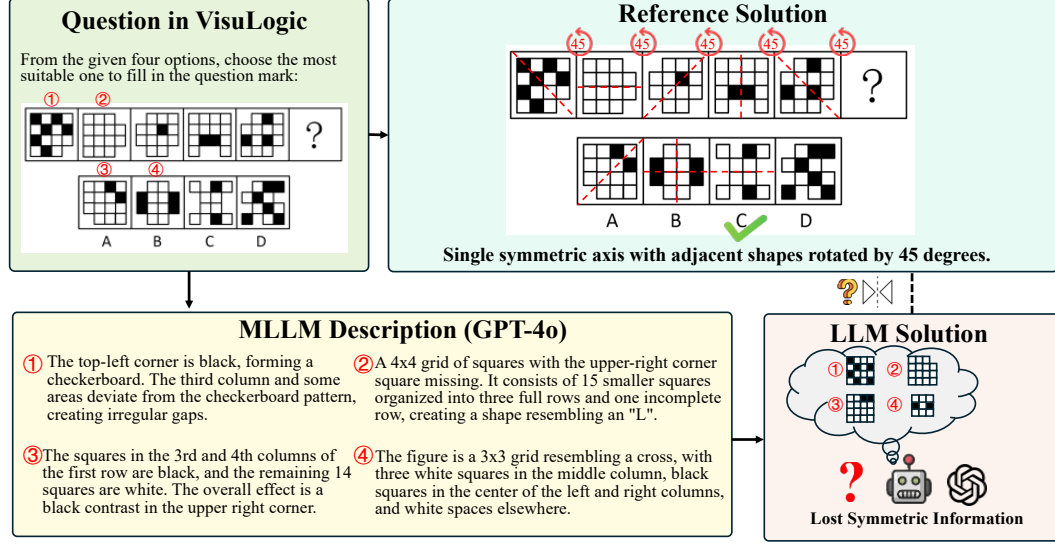


Figure 1: **Composition of the VisuLogic benchmark and performance of representative MLLMs.** The left figure shows the distribution of the 6 categories and their subcategories in VisuLogic. The right figure shows accuracies (%) achieved by MLLMs and by human on each category of VisuLogic.

equal contribution; [†] interns at OpenGVLab, Shanghai AI Laboratory; ^{*} corresponding author.



(a) Pipeline of “MLLM description → LLM” for Question in MMMU [89]. It is trivial that SOTA MLLMs extract key visual details, thereby enabling the LLM to answer questions solely based on language reasoning.



(b) Pipeline of “MLLM description → LLM” for Question in VisuLogic. Even SOTA MLLMs struggle to describe images precisely, leading to ambiguous interpretations.

Figure 2: Comparison of the “MLLM description → LLM” pipeline on two benchmarks. In MMMU, detailed descriptions lead to correct solutions, while in VisuLogic, critical visual cues (e.g., symmetry, rotation) can be easily lost, causing the LLM to misinterpret the image. This highlights that textual reasoning alone is insufficient, underscoring the benchmark’s demand for robust and in-depth visual reasoning.

Reasoning, as fundamental component of human intelligence, has become a critical criterion in evaluating progress toward Artificial General Intelligence (AGI) [26, 74]. Recent advancements in Large Language Models (LLMs) have demonstrated substantial improvements in reasoning capabilities across complex domains such as mathematics [61, 82, 81, 58], logical reasoning [68, 79, 23, 47] and coding [2, 35, 42, 32]. Techniques like Chain-of-Thought (CoT) [75] prompting and test-time compute scaling (e.g., OpenAI o1 [34] and Deepseek-R1 [18]) have significantly enhanced the reasoning performance of LLMs [18, 26, 74]. Along with the rapid development of language reasoning research for LLMs, considerable progress [84, 61, 58, 11, 50, 72, 51, 63, 73] has been made in improving multimodal reasoning capability of Multimodal Large Language Models (MLLMs).

These methods, which often incorporate reinforcement learning techniques [11, 50, 61] to enhance the reasoning capabilities of MLLMs, have achieved some early successes [84, 61, 58, 11, 50, 51, 63]. However, they typically rely on existing multi-modal benchmarks that struggle to accurately capture a model’s core visual reasoning ability. For example, VLM-R1 [63] assesses “visual reasoning” with referring expression comprehension tasks [88, 55, 38], yet these tasks primarily focus on object localization, demanding only basic perceptual skills rather than more advanced visual cognitive processes. Meanwhile, several works [58, 61, 84] adopt mathematical problem-solving benchmarks that include diagrams—such as MathVista [52], MathVerse [91], and MathVision [69]—to evaluate visual reasoning. In practice, however, as [91] observes, many MLLMs translate these visual clues into textual descriptions and then rely on standard language reasoning. This approach can incorrectly

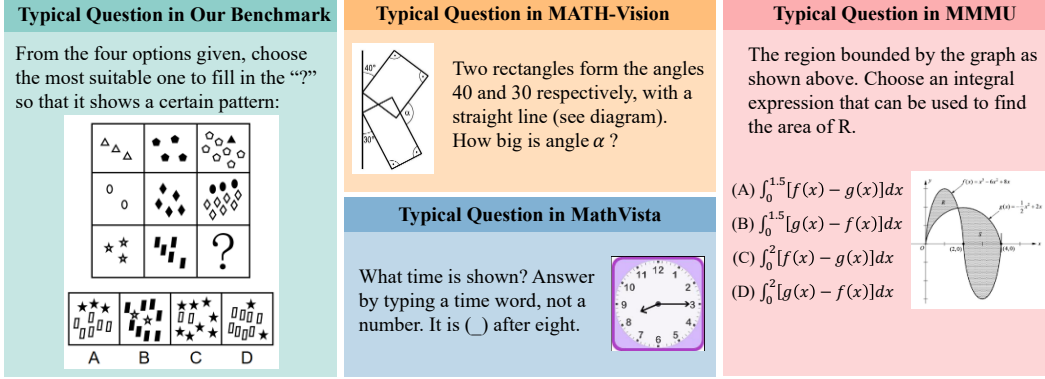


Figure 3: **Comparison of questions from different Benchmarks.** Compared to MathVista [52], MathVision [69], and MMMU [89], VisuLogic focuses more explicitly on assessing pure visual reasoning capabilities.

attribute language-driven results to visual reasoning, resulting in a misleading assessment of the model’s visual reasoning capabilities [91, 30]. Consequently, designing new benchmarks that explicitly focus on vision-centric reasoning—rather than conflating it with text-based reasoning—remains critical for advancing MLLMs’ visual reasoning capabilities.

To address this limitation, we propose VisuLogic, a novel benchmark specifically designed to evaluate visual reasoning abilities in multimodal models without mixing them with purely text-based reasoning (see Figure 3). VisuLogic comprises carefully constructed tasks that span multiple reasoning categories (see Figure 1). As shown in Figure 5, these tasks are classified into six key types, such as Quantitative Reasoning, which requires understanding and deducing shifts in the quantity of certain elements within an image. In contrast to existing benchmarks, as demonstrated in Figure 2, state-of-the-art (SOTA) MLLMs often omit crucial visual details when describing VisuLogic problems, making it difficult for them to rely solely on a text-based inference shortcut. Indeed, even humans would find it challenging to capture every essential visual cue in a single description, so effectively tackling VisuLogic demands more robust, vision-centric reasoning. By reducing reliance on textual inference shortcuts, VisuLogic thus provides a stringent evaluation of MLLMs’ genuine visual reasoning capabilities.

We conducted a comprehensive evaluation and systematic analysis to assess current models’ visual reasoning capabilities. When leading text-only LLMs were supplied with detailed descriptions in place of raw images, their accuracy—Doubao-1.5-Pro (26.6%), Claude-3.7-Sonnet (25.9%) and Qwen2.5-72B-Instruct [83] (28.0%)—barely exceeded the random-chance baseline of 24.9%. This clearly demonstrates that textual reasoning alone are insufficient for solving our VisuLogic tasks. Even state-of-the-art multimodal large language models (MLLMs)—including GPT-4o [33], Doubao-1.5-Vision-Pro, Gemini-2.0-Pro-Exp [64] and InternVL3-78B [94]—achieve only 26.3%, 28.1%, 28.0% and 27.7%, respectively, whereas human participants reached 51.4%. The substantial gap between these results and human performance underscores the challenge of robust visual reasoning in current MLLMs. Furthermore, we applied a simple reinforcement-learning (RL) fine-tuning step on our supplementary training dataset: this boosted the baseline model’s accuracy from 25.5% to 31.1%, outperforming both open-source and closed-source counterparts. These findings illustrate the promise of the RL technique for strengthening MLLMs’ visual reasoning capabilities.

In summary, our contributions are as follows:

- We propose a challenging visual reasoning benchmark that is inherently difficult to articulate using language, providing a more rigorous evaluation of the visual reasoning capabilities of MLLMs.
- We conduct comprehensive experiments to evaluate and analyze the benchmark, including extensive evaluations and comparative studies of various MLLMs under different setting.
- We identify the RL technique as a promising direction for improving the visual reasoning capabilities of MLLMs. Furthermore, we release both the training code and data to facilitate future research.

2 Related Work

Multi-modal Large Language Models. Recent years have witnessed substantial advancements in Multi-modal Large Language Models (MLLMs). Early works like BLIP [41, 40] and Flamingo [5] introduce lightweight parameters between vision transformer [21] (ViT) and LLMs, laying the groundwork for multimodal perception. Subsequent efforts, such as LLaVA [45] and MiniGPT-4 [93], integrate instruction tuning, further enhancing the performance of MLLMs. Proprietary models like GPT-4o [33] and Gemini-Pro [64] have advanced MLLM performance on complex multimodal tasks, while open-source models such as Qwen-VL series [7, 70, 8] and InternVL series [15, 16, 24, 14, 94] achieve competitive results through optimized architectural design, dataset expansion and training paradigm improvements. Meanwhile, some related studies further advance the ability of large models by incorporating new modalities (e.g., audio [22, 19, 77], point clouds [27, 9], video [92, 12]) and by supporting more tasks (e.g., grounding [80, 71], computer usage [60, 6]). Notably, limited research attempts to enhance the reasoning capabilities of MLLMs. Some pioneering works, such as R1-Onevision [84], LMM-R1 [61], MM-EUREKA [58], R1-V [11], Visual-rft [50], Visualprm [72], OThink-MR1 [51], VLM-R1 [63], and Open-r1-Video [73] have explored the visual reasoning capabilities of MLLMs through Reinforcement Learning (RL), but they are still in the nascent stage.

Multimodal Benchmarks. With the development of MLLMs, multimodal benchmarks have also evolved significantly [43]. Early benchmarks primarily address visual perception tasks through simple tasks like visual question answering (VQA) [13, 44, 36, 78], image captioning [59, 20, 37] and referring expression comprehension [88, 55]. Subsequent works expand the capability coverage of benchmarks into more specialized domains: OCRBench [49], Chartqa [56] and DocVQA [57] assess textual content extraction; AgentBench [48] and ToolEyes [86] test tool usage capabilities; and egocentric perception benchmarks [54, 17] quantify first-person scene interpretation. Despite the progress, they ignore the evaluation of visual reasoning abilities [90, 89]. Recently, some benchmarks have made explorations in examining MLLMs’ visual reasoning abilities, but methodological deficiencies still cause limitations to assess the intrinsic visual reasoning capabilities [30, 4, 76]. InfIMM-Eval [29] test reasoning abilities around daily life, lacking deep-level reasoning scenarios. MMMU [89] and Emma [30] provide benchmarks demanding advanced reasoning abilities in fields such as chemistry and physics, but they ignore questions around the images’ fundamental visual components (e.g., shapes, elements). While mathematical benchmarks [69, 52, 31, 62, 91, 28] evaluate mathematical reasoning with geometric and diagram problems included, they focus on math capabilities but disregard logical analysis about the vision information. LogicVista [76] provides a multimodal logical reasoning benchmark, its visual questions lack analytical depth—dominated by single-hop, superficial queries in limited data scope. Unlike previous works, we introduce a challenging benchmark focused specifically on the domain of visual logical reasoning.

3 VisuLogic

In this section, we first describe the VisuLogic data-curation pipeline, which comprises three key stages: data collection, quality control, and the detailed taxonomy. We then report the benchmark’s construction statistics, including total size, answer-option distributions, and category-level proportions. Finally, we introduce a supplementary training dataset—consisting of questions analogous to those in VisuLogic—designed to bolster future research and facilitate community engagement.

3.1 Data Curation Pipeline

Data Collection. We construct the VisuLogic dataset by sourcing all questions from publicly available online resources in compliance with relevant licenses and regulations. As shown in Figure 4, our automated data processing pipeline comprises three stages: 1) **Fetching:** We employ Playwright¹ to systematically scrape raw web content, supplemented by custom parsing scripts that extract question-answer pairs. 2) **Cleaning:** We remove noise, irrelevant content, and extraneous HTML markup (e.g., `<div>`) to ensure the integrity of the textual data. 3) **Structuring:** We standardize the cleaned text and images by structuring all information (such as question text, metadata) in JSON Lines (JSONL) format.

¹<https://github.com/microsoft/playwright>

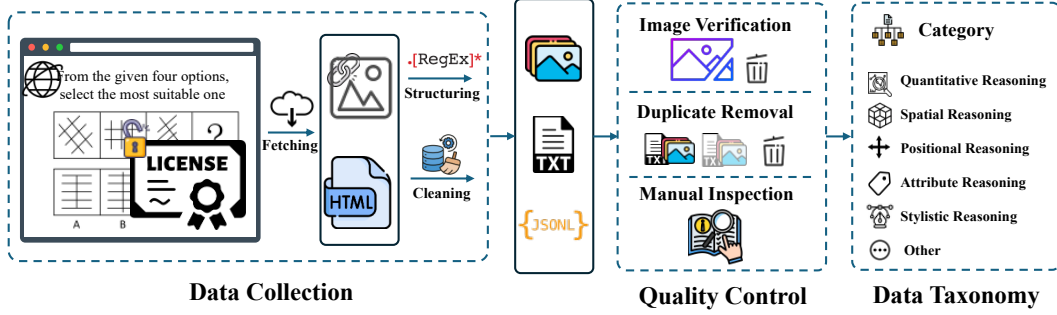


Figure 4: **Data curation pipeline of VisuLogic.** The pipeline includes Data Collection, Quality Control and Data Taxonomy.

Quality Control. To ensure the reliability of the benchmark dataset, we employ a three-stage data validation procedure: 1) **Image Verification:** Each image referenced in the questions is checked for existence and correct formatting; any item that fails to meet the criteria is removed following human review. 2) **Duplicate Removal:** We eliminate redundant entries at both the text and image levels by (i) detecting lexical overlap among text segments and (ii) applying perceptual hashing (pHash) to identify visually similar images. 3) **Manual Checking:** After automated filtering, we perform a thorough human-led review of every remaining entry to confirm its validity and ensure dataset reliability.

Data Taxonomy. We categorize all collected data into a taxonomy of six primary classes based on expert human annotation of the reasoning skills each question requires. Annotators first tag questions according to the targeted reasoning competency; these annotated tags are then analyzed and merged into five primary categories. A subsequent human review ensures that every question is accurately classified, with any ambiguous instances consolidated under the “Other” category. Specifically, we define each category as follows. **Quantitative Reasoning** focuses on changes in the number or count of graphical elements (for example, points, lines and angles) and on arithmetic relationships among shapes. **Spatial Reasoning** requires mentally reconstructing three-dimensional shapes from two-dimensional figures, folding or unfolding surfaces, and integrating three-dimensional structures. **Positional Reasoning** examines transformations such as translation, rotation and reflection of objects while preserving their fundamental elements. **Attribute Reasoning** involves intrinsic properties of shapes, including symmetry (axial or central), curvature and measures of openness or closedness. **Stylistic Reasoning** entails alterations in stylistic features such as overlay, subtraction and assessments of shape similarity or difference. **Other** encompasses questions that fall outside the preceding categories, including those involving letters, alphanumeric symbols or other specialized characters.

3.2 Dataset Statistics

Following data curation and validation, VisuLogic comprises 1,000 single-choice questions. Figure 1 (left) illustrates the category distribution: Quantitative Reasoning (35.3%), Spatial Reasoning (23.1%), Positional Reasoning (13.6%), Attribute Reasoning (8.2%), Stylistic Reasoning (9.0%), and Other (10.8%). Correct answer options are evenly balanced, with the proportions distributed as follows: A (23.1%), B (26.7%), C (25.2%), and D (25.0%).

3.3 Supplementary Training Dataset

To facilitate further investigation of visual reasoning, we provide an auxiliary training set of 4,296 question-answer pairs drawn from the same domains and subjected to identical validation procedures to prevent overlap with the benchmark. The training split mirrors the primary taxonomy, with category proportions of Quantitative Reasoning (30.7%), Spatial Reasoning (25.5%), Positional Reasoning (13.0%), Attribute Reasoning (8.8%), Stylistic Reasoning (9.9%), and Other (12.1%).

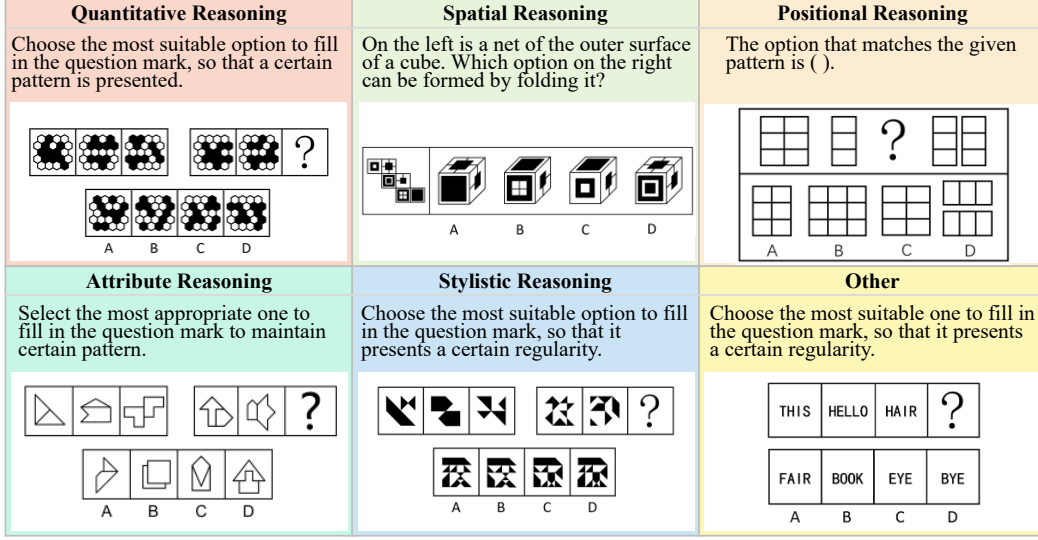


Figure 5: **Question examples of different categories in our VisuLogic Benchmark.** VisuLogic contains 6 categories of questions, which require models’ abilities in visual logic reasoning.

4 Experiments

In this section, we present a comprehensive evaluation of the VisuLogic benchmark. We first describe the experimental setup in Section 4.1, followed by overall performance results in Section 4.2. We then analyze systematic errors in Section 4.3 and provide qualitative insights in Section 4.4.

4.1 Experiment Setup

References Performance. To fully investigate models’ performance, we establish two reference points: 1) **Human Performance:** We invited 100 graduate students majoring in science and engineering to solve 10 randomly sampled VisuLogic questions each, allowing 2–5 minutes per question. The aggregate accuracy over all participants constitutes the human benchmark. 2) **Random Selection:** We simulate random guessing by sampling answers uniformly over 10 independent runs and report the average accuracy as the random baseline.

Evaluated Models. We evaluate a total of 28 models on VisuLogic, comprising 8 large language models (LLMs) and 20 multimodal large language models (MLLMs). For open-source LLMs, we test *Deepseek-R1* [18], *Qwen2.5-72B-Instruct* [83] and *Qwen-QwQ* [67], and for close-source LLMs we evaluate *GPT-4* [1], *o3-mini*, *Gemini-2.0-Flash-Thinking* [64], *Claude-3.7-Sonnet* and *Doubao-1.5-Pro-32k*. Open-source MLLMs include *Qwen2.5-VL-7B-Instruct* [8], *Qwen2.5-VL-72B-Instruct* [8], *QvQ-72B-Preview* [66], *InternVL2.5-38B* [16], *InternVL2.5-78B* [16], *InternVL3-38B* [94], *InternVL3-78B* [94], *LLaV -v1.5-7B* [46], *LLaV -OneVision-7B (SI)* [39], *ShareGPT4V* [10], *MiniCPM-o-2.6* [85], *GLM-4v-9B* [25], *Ovis2-8B* [53] and *mPLUG-Owl3-7B* [87], while close-source MLLMs consist of *GPT-4o* [33], *GPT-4o-mini*, *Kimi-latest* [65], *Doubao-1.5-Vision-Pro-32k*, *Gemini-2.0-Pro* [64] and *Claude-3.7-Sonnet*. We further include two reinforcement-learning baselines built on *Qwen2.5-VL-7B-Instruct* [8] and *InternVL2.5-38B* [16], respectively, trained via our rule-based RL procedure on our supplementary training dataset. Fully supervised fine-tuning (SFT) experiments on the same datasets serve as controls to isolate the effect of RL optimization. All model hyperparameters, training regimes, and implementation details are provided in the Appendix.

LLM Evaluation Protocol. For language-only models, we generate an auxiliary image description using GPT-4o and prepend it to the question. Specifically, each question is formatted as “*Following is a detailed caption describing an image: [IM GE DESCRIPTION]. Based on the provided description, select the best answer from the four options:*”. This combined prompt is fed directly into the target LLMs for inference.

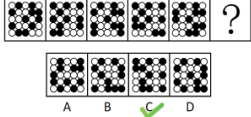
Problem From the four options given, choose the most appropriate one to fill in the question mark so that it shows a certain regularity.  A B C D a	GPT-4o ✗ To determine the correct answer, we need to identify the pattern in the sequence of images. Upon analyzing the sequence, each figure shows a progression where black and white dots alternate positions in a specific and regular way. Observations: 1. In each step, the black dots shift positions, creating a systematic pattern. 2. The white and black dots alternate in a way that spreads uniformly across the grid. By analyzing the progression in the sequence of grids, Option A is the correct choice as it aligns with the observed regularity. e
Reference Solution The elements are the same, so the position pattern is given priority. The figures in the question all have 25 squares, and the number of black blocks in the middle is the same. So the inner and outer circles are considered separately for translation. It is found that the position of the black block in the center of the inner circle remains unchanged, and the other two black blocks move 4 squares clockwise (counterclockwise) each time (it can also be seen as the positions of the two black blocks alternating on the left and right sides), and the outermost black block moves 1 square clockwise each time. Only option C is correct. b	QwenVL2.5 72B-Instruct ✗ To determine the most appropriate option to fill in the question mark, we need to identify the pattern in the sequence of grids. Let's analyze the given grids step by step. The first grid has 10 black circles and 6 white circles. The second grid has 9 black circles and 7 white circles. The third grid has 8 black circles and 8 white circles. We can observe that the number of black circles decreases by 1 and the number of white circles increases by 1 in each subsequent grid. Therefore, the fourth grid should have 7 black circles and 9 white circles. Now, let's check the options: A: This grid has 7 black circles and 9 white circles. B: This grid has 6 black circles and 10 white circles. C: This grid has 8 black circles and 8 white circles. D: This grid has 9 black circles and 7 white circles. The grid that matches our pattern is option A, which has 7 black circles and 9 white circles. Thus, the most appropriate option to fill in the question mark is <code>\boxed{A}</code>. f
InternVL2.5 38B ✗ According to the changing pattern of these graphs, we can observe that: The number of black circles in each graph is gradually decreasing, which allows us to analyze the options in the last row. Let's look at each option: - The number of black circles in options A and B is more than in the previous graphs, which is obviously inconsistent with the pattern. - There are more black circles in option C than before, which is inconsistent with the previous pattern. - The number of black circles in option D is consistent with the previous change pattern. - Therefore, option D is the most suitable option. c	InternVL2.5 38B + RL ✓ To answer this question, we need to carefully observe the sequence of figures and find the pattern between them. First, let's look at the position and changes of the circles in each figure. The number of circles in each figure is the same, but their positions seem to be changing in some way. After careful analysis, we can see that the positions of the circles in the figures are moving in a certain order. 1. In the first figure, the positions of the circles are relatively concentrated. 2. In the second figure, the positions of the circles change slightly and move slightly more than the first figure. 3. In the third figure, the positions of the circles change again and continue to move according to a certain pattern. 4. In the fourth figure, the positions of the circles are closer to the center. 5. In the fifth figure, the positions of the circles continue to move compared to the fourth figure. 6. In the sixth figure, the positions of the circles are different from the other figures, forming a new arrangement. Through the analysis of the above figures, we can see that the current pattern of changes in the figures seems to be gradually forming a new arrangement. Therefore, the seventh figure should continue this change and form a new arrangement. According to this rule, the arrangement of option C is most consistent with the arrangement of the next figure. So, the answer is: C: C g
Image Description (GPT-4o) Structure of the Puzzle: • The top row consists of five grids, each with a 6x6 arrangement of black and white circles. • The first four grids follow a specific pattern, while the fifth grid is missing (represented by a "?"). • The goal is to determine the pattern and select the correct option from four answer choices (A, B, C, D) shown below. Observations About the Given Sequence: 1. Each grid in the sequence contains a mix of black and white circles. 2. The arrangement of black circles seems to follow a transformation from one grid to the next. 3. By analyzing the first four grids, a rule or pattern governing the change can be identified. Answer Choices (A, B, C, D): Each of the four answer choices consists of a 6x6 grid similar to the ones in the sequence. Only one of them correctly follows the pattern observed in the first four grids. d	o3-mini ✗ (Reasoned for 5 seconds.) To solve this puzzle, let's follow a step-by-step reasoning approach based on the observations about the sequence of grids. 1. **First Step: Observe the Pattern in the First Four Grids** - We need to closely examine the black and white circles in each grid, paying attention to how the black circles evolve across each grid. - Look at the number of black circles, their positions, and any shifts or transformations from one grid to the next. 2. **Identify Possible Transformations** - The transformation could be based on the movement of black circles, changes in the arrangement (e.g., rotation, mirroring, shifting), or adding/removing black circles in a predictable manner. - Consider changes like: - Moving black circles from one position to another. - Adding or removing black circles. - Symmetry or patterns like shifting rows or columns. 3. **Predict the Missing Grid**- Using the transformations observed from the first four grids, hypothesize what the fifth grid should look like. - Think about the positions, number of black circles, and their movements. The fifth grid should be the natural progression based on the previous grids. 4. **Compare the Answer Choices** - Once you have a hypothesis for the missing grid, compare it against the four answer choices (A, B, C, D). - Choose the one that matches the predicted transformation. Based on this reasoning approach, let's assume that you've carefully analyzed the first four grids and identified a clear pattern in the movement and arrangement of the black circles. Answer: \boxed{B}. h

Figure 6: **Solution examples generated by different models.** Reference solution and outputs generated by GPT-4o [33], Qwen2.5VL-72B-Instruct [8], InternVL2.5-38B [16], and InternVL2.5-38B with RL. Additionally, the image description and solution from LLMs (o3-mini) are also illustrated.

Prompts Setting. We apply three distinct prompting paradigms to investigate model reasoning capabilities: 1) **Non-CoT prompt evaluation:** Models receive a concise instruction: “*answer the question using a single word or phrase, following this format: answer: \boxed{\$LETTER}*”. 2) **CoT prompt evaluation:** We prompt models to articulate intermediate reasoning steps: “*Solve the complex visual logical reasoning problem through step-by-step reasoning. Think about the reasoning process first and answer the question following this format: answer: \boxed{\$LETTER}*”. 3) **Hint prompts evaluation:** Leveraging GPT-4o, we generate question-specific hints derived from the reference solutions. As shown in Figure 7, solution-related hints are provided alongside the CoT prompt to guide reasoning without revealing the final answer directly. Notably, unless otherwise specified, CoT prompt evaluation is employed by default for assessing model performance.

4.2 Overall Results

LLM Performance. Table 1 reports that all evaluated LLMs attain rather low accuracy on VisuLogic. The best-performing LLM, *Qwen2.5-72B-Instruct*, reaches only 28.0%, while *GPT-4* and *Deepseek-R1* achieve 23.6% and 26.6%, respectively. These findings underscore that reasoning based solely on

Table 1: **Cross-Modal performance with CoT prompts on VisuLogic.** The table shows the evaluation scores of baseline references, LLMs, and MLLMs, which illustrates a gap between humans’ and models’ capabilities. Top performers per category are **bolded**, with secondary leaders underlined.

Models	Overall	Quantity	Spatiality	Position	Attribute	Style	Other
References							
Human	51.4	45.3	52.7	71.1	50.0	47.5	44.2
Random	24.9	25.7	25.4	22.7	23.4	24.3	26.1
Open Source LLM (MLLM Description→LLM)							
Deepseek-R1 [18]	<u>26.6</u>	<u>27.7</u>	<u>23.5</u>	24.0	27.8	<u>23.0</u>	35.0
Qwen2.5-72B-Instruct [83]	28.0	30.2	24.4	27.5	<u>26.5</u>	26.8	<u>30.8</u>
QwQ-32B [67]	22.8	24.6	20.1	<u>25.4</u>	19.0	20.7	24.0
Close Source LLM (MLLM Description→LLM)							
GPT-4 (20240613) [1]	23.6	21.2	<u>22.5</u>	21.3	<u>25.6</u>	23.3	35.2
o3-mini (20250131)	24.6	<u>27.8</u>	18.8	24.5	21.7	<u>25.6</u>	28.4
Gemini-2.0-Flash-Thinking (20250121) [64]	23.4	23.2	26.0	16.9	17.1	21.1	<u>33.3</u>
Claude-3.7-Sonnet (20250219)	<u>25.9</u>	26.6	<u>22.5</u>	25.0	28.0	<u>25.6</u>	30.6
Doubao-1.5-Pro-32k (20250115)	26.6	30.0	<u>22.5</u>	25.0	<u>25.6</u>	30.0	24.1
Close Source MLLMs							
GPT-4o-mini (20240718)	24.3	27.2	23.4	23.5	18.3	<u>31.1</u>	16.7
GPT-4o (20240806) [33]	26.3	<u>28.6</u>	24.7	27.2	26.8	20.0	25.9
Kimi-latest [65]	25.9	24.9	29.4	26.5	<u>28.0</u>	16.7	26.9
Doubao-1.5-Vision-Pro-32k (20250115)	28.1	28.1	23.8	29.1	25.1	32.1	35.0
Gemini-2.0-Pro (20250205) [64]	<u>28.0</u>	29.7	24.2	<u>27.9</u>	30.5	22.2	<u>33.3</u>
Claude-3.7-Sonnet (20250219)	24.8	22.7	<u>27.3</u>	<u>27.9</u>	<u>28.0</u>	22.2	22.2
Open Source MLLMs							
LLaVA-v1.5-7B [46]	24.6	26.1	24.2	23.5	17.1	<u>31.1</u>	22.2
LLaVA-OneVision-7B (SI) [39]	25.3	22.4	<u>27.3</u>	33.1	23.2	25.6	22.2
ShareGPT4V [10]	23.4	24.9	22.1	23.5	19.5	28.9	19.4
MiniCPM-o-2.6 [85]	25.3	25.6	23.0	27.3	21.9	24.5	29.9
GLM-4v-9B [25]	24.3	22.4	23.7	28.3	26.0	24.1	25.3
Ovis2-8B [53]	25.6	26.1	23.8	27.2	28.0	25.6	24.1
mPLUG-Owl3-7B-241101 [87]	18.9	21.5	15.2	16.2	20.7	18.9	20.4
Qwen2.5-VL-7B-Instruct [8]	26.0	27.6	20.9	25.2	23.2	37.8	25.0
Qwen2.5VL-72B-Instruct [8]	26.2	25.2	23.8	27.2	25.6	25.6	34.3
QvQ-72B-Preview [66]	23.0	24.2	17.0	24.4	21.0	24.4	30.6
InternVL2.5-38B [14]	25.5	24.4	26.4	27.2	23.2	25.6	26.9
InternVL2.5-78B [14]	<u>27.3</u>	26.6	26.0	26.5	<u>26.8</u>	<u>31.1</u>	30.6
InternVL3-38B [94]	27.1	28.7	27.6	26.1	21.4	23.9	28.5
InternVL3-78B [94]	27.7	<u>27.7</u>	26.1	<u>31.6</u>	26.3	21.3	<u>32.3</u>
Qwen2.5-VL-7B-Instruct-SFT	25.5	24.4	26.4	<u>27.2</u>	23.2	25.6	26.9
Qwen2.5-VL-7B-Instruct-RL	<u>28.0</u>	26.6	33.8	29.4	23.2	18.9	29.6
InternVL2.5-38B-SFT	27.9	<u>30.6</u>	29.4	20.6	<u>25.6</u>	30.0	25.0
InternVL2.5-38B-RL	31.1	31.2	<u>31.2</u>	26.5	30.5	30.0	38.9

textual descriptions is insufficient to capture the rich visual information required by our benchmark, causing failures to resolve visual logical reasoning problems.

MLLM Performance. As shown in Table 1, current multimodal LLMs also perform poorly on VisuLogic. The highest score is 28.1% by *Doubao-1.5-Vision-Pro-32k*, which remains a substantial 23.3 points below human performance. Advanced models such as *GPT-4o* and *Gemini-2.0-Pro* attain only 26.3% and 28.0%, respectively, revealing a marked gap between existing MLLMs and human-level visual reasoning. Overall, these results indicate that current MLLMs have serious deficiencies in visual reasoning and that significant advances are still required.

Effectiveness of CoT Prompts. Contrary to expectations, chain-of-thought (CoT) prompting yields minimal improvements in visual reasoning. As detailed in Table 2, *GPT-4o-mini* benefits most, with only a 1.2-point gain under CoT compared to direct-answer prompts; all other models exhibit gains below 1.0 point. We speculate that this limited effect likely stems from current CoT training being

Table 2: **Influence of Chain-of-Thought on model performance.** Positive value changes are highlighted in **red**, negative changes in **green**, and statistically insignificant variations (delta < 1%) are denoted in **gray**. With CoT prompts, MLLMs only exhibit tiny improvements in visual reasoning.

Models	CoT	Overall	Quantity	Spatiality	Position	Attribute	Style	Other
GPT-4o (20240806)	✓ ✗	26.3 26 0 3)	28.6 26 9 1 7)	24.7 24 2 0 5)	27.2 26 5 0 7)	26.8 23 2 3 6)	20.0 24 +4 0)	25.9 29 6 +3 7)
Kimi-latest	✓ ✗	25.9 25 1 0 8)	24.9 22 9 2 0)	29.4 22 5 6 9)	26.5 25 1 5)	28.0 19 5 7 5)	16.7 35 6 +18 9)	26.9 24 1 2 8)
GPT-4o-mini (20240718)	✓ ✗	24.3 23 1 1 2)	27.2 23 8 3 4)	23.4 22 9 0 5)	23.5 24 3 +0 8)	18.3 17 1 1 2)	31.1 3 1 1)	16.7 18 5 +1 8)
Qwen2.5-VL-Instruct-7B	✓ ✗	26.0 25 9 0 1)	27.6 25 5 2 1)	20.9 22 8 +1 9)	25.2 26 4 +1 2)	23.2 25 3 +2 1)	37.8 2 6 17 2)	25.0 38 2 +13 2)
InternVL2.5-38B	✓ ✗	24.9 25 +0 1)	24.1 24 6 +0 5)	26.4 25 5 0 9)	27.2 22 1 5 1)	23.2 22 1 2)	25.6 26 7 +1 1)	22.2 29 6 +7 4)

Table 3: **Influence of hint prompts on model performance.** MLLMs exhibit measurable performance enhancements with hint integration, yet retain significant gaps against human performance. In comparison, humans achieve task mastery on VisuLogic with hints. Value changes are color-coded with **red** indicating positive shifts and **green** denoting negative variations.

Models	Hint	Overall	Quantity	Spatiality	Position	Attribute	Style	Other
Human	✗ ✓	51.4 83 6 +32 2)	45.3 85 1 +39 8)	52.7 68 5 +15 8)	71.1 1 +28 9)	50.0 95 7 +45 7)	47.5 78 6 +31 1)	44.2 9 5 +46 3)
GPT-4o (20240806)	✗ ✓	26.3 3 +3 7)	28.6 25 4 3 2)	24.7 31 5 +6 8)	27.2 29 2 +2 0)	26.8 28 6 +1 8)	20.0 3 8 +10 8)	25.9 42 9 +17 0)
Claude-3.7-Sonnet (20250219)	✗ ✓	24.8 33 5 +8 7)	22.7 37 3 +14 6)	27.3 33 3 +6 0)	27.9 37 5 +9 6)	28.0 23 8 4 2)	22.2 15 4 6 8)	22.2 38 1 +15 9)
Gemini-2.0-Pro (20250205)	✗ ✓	28.0 36 5 +8 5)	29.7 44 8 +15 1)	24.2 33 3 +9 1)	27.9 25 2 9)	30.5 38 1 +7 6)	22.2 15 4 6 8)	33.3 42 9 +9 6)
Doubao-1.5-Vision-Pro-32k (20250115)	✗ ✓	28.1 37 +8 9)	28.1 46 3 +18 2)	23.8 25 9 +2 1)	29.1 54 2 +25 1)	25.1 33 3 +8 2)	32.1 23 1 9 0)	35.0 28 6 6 4)

based only on pure-text corpora; future works should explore CoT techniques tailored to multimodal data to better support visual reasoning tasks.

Effectiveness of Hint Prompts. Table 3 shows that hint prompts can boost model performance—*Claude-3.7-Sonnet*, *Gemini-2.0-Pro*, and *Doubao-1.5-Vision-Pro-32k* all improve by over 8 points, reaching accuracies above 35%. However, even with explicit guidance, models still fail to construct coherent, reliable reasoning chains. This suggests that simply augmenting training data with similar tasks is insufficient (which can help MLLMs come up with specific directions for solving the problem); future efforts must focus on enhancing the reliability and correctness of reasoning procedures of MLLMs to achieve more accurate reasoning inference.

Impact of Model Scaling. In Table 1, we observe a positive correlation between parameter size and model performance. Within the same model series, *Qwen2.5-VL-72B-Instruct* achieves 26.2% outperforming *Qwen2.5-VL-7B-Instruct* (26.0%) by 0.2%. Furthermore, *InternVL2.5-78B* (27.3%) surpasses *InternVL2.5-38B* (25.5%) by a margin of 1.8%.

Open-Source vs Close-Source. Table 1 further compares open- and closed-source models. The top open-source MLLM, *InternVL3-78B*, attains 27.7%, trailing the closed-source leader (*Doubao-1.5-Vision-Pro-32k*, 28.1%) by only 0.4% points and outperforming other proprietary competitors such as *GPT-4o* and *Claude-3.7-Sonnet*. Overall, both open- and closed-source models exhibit uniformly low performance, highlighting a widespread neglect of visual reasoning objectives in current multimodal model training and data collection.

Behaviors of RL Trained models. As shown in Table 1, MLLMs with reinforcement learning optimization can yield obvious improvements in visual reasoning performance. *Qwen2.5-VL-7B-Instruct-RL* attains 28.0%, a 2.0 percentage point boost over its non-RL counterpart. More strikingly, *InternVL2.5-38B-RL* reaches 31.1%, surpassing the original non-RL model by 5.6% and establishing a new state-of-the-art on VisuLogic. Furthermore, compared to supervised fine-tuning (SFT) on iden-

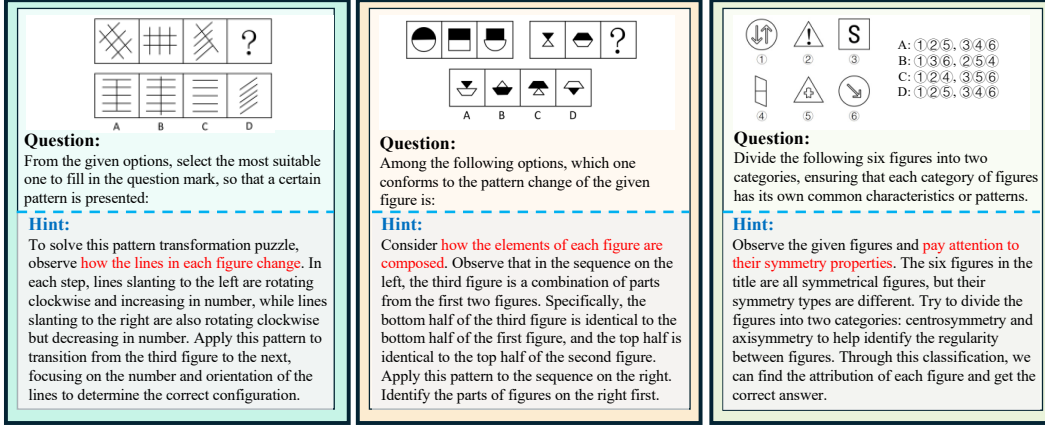


Figure 7: **Hint prompts visualization.** Hint prompts examples, which supply solution guidance for MLLMs, are shown in the image, with solution-critical elements highlighted in red.

tical datasets, RL-enhanced models demonstrate substantially larger performance gains, underscoring the promise of targeted RL methods for advancing multimodal visual reasoning.

4.3 Fine-grained Comparison

We systematically analyze model capabilities by examining error distributions across reasoning categories for different models. Figure 8 presents the error rates of LLMs, MLLMs, and human participants over six distinct reasoning categories.

Figure 8a reveals that LLMs struggle most with *Spatial Reasoning* questions, indicating that text-only descriptions are insufficient to infer three-dimensional structures or spatial transformations. In contrast, their performance on *Quantitative Reasoning* tasks is comparatively stronger, suggesting that quantitative relationships are more readily conveyed through language.

As shown in Figure 8b, *Stylistic Reasoning* presents the greatest difficulty for MLLMs, with error rates exceeding 75%—worse than random guessing (25% accuracy). This result underscores a fundamental limitation of current MLLM architectures in capturing subtle visual cues such as overlays, contours, and shape variations.

Figure 8c reveals that human error patterns form a distinct cluster, separate from both LLMs and MLLMs. Human participants maintain error rates below 30% on *Positional Reasoning* tasks, reflecting robust position-based visual inference. By contrast, both model classes struggle with positional reasoning, highlighting a fundamental divergence in visual-cognitive processes between humans and MLLMs.

4.4 Qualitative analysis

LLM Failures. As shown in Figure 6(h), text-only LLMs that rely on externally generated image captions often omit critical visual details required for multi-step logical deduction—such as the counts, shapes, and progression patterns of the black and white dots in Figure 6(a). Consequently, their reasoning diverges from the correct solution and frequently yields hallucinations or irrelevant responses.

MLLM Failures. Figure 6 also presents cases in which MLLMs correctly describe static visual content yet fail to infer the evolving relationships among shapes, instead resorting to superficial cues like object counts. While these models can recognize individual shapes and tally items, they struggle to reason over inter-element relations, which limits their ability to solve complex visual-logic problems.

RL-Based Improvements. As illustrated in Figure 6(g), reinforcement learning (RL) encourages deeper, stepwise logical reasoning. The RL-enhanced model successfully captures state transitions (e.g., the movements of chess pieces in Figure 6(a)) and accurately predicts subsequent configurations. Moreover, it learns to iteratively revise intermediate hypotheses—akin to trial-and-error—until a

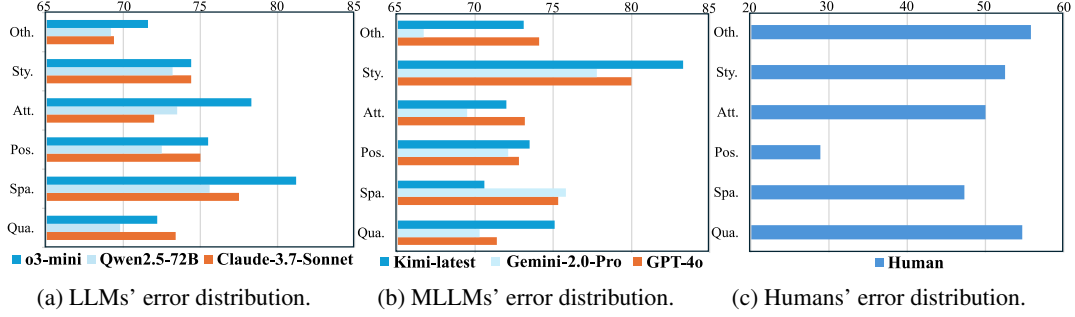


Figure 8: **Error distribution analysis.** The figure demonstrates distinct error type allocations across Humans, LLMs and MLLMs, revealing differences among their cognition patterns.

coherent deduction emerges (see additional examples in the Appendix). These findings highlight the potential of RL methods to bolster performance on visual reasoning tasks.

5 Conclusion

In this paper, we present VisuLogic, a novel benchmark designed to evaluate the visual reasoning capabilities of Multi-modal Large Language Models (MLLMs). The benchmark consists of 1,000 vision-centric reasoning tasks distributed across six distinct categories. We conduct comprehensive evaluation of several advanced LLMs and MLLMs on this benchmark and provide an in-depth analysis of their performance. Our findings reveal that even the most advanced models fall short of human performance, highlighting substantial opportunities for advancement in visual logical reasoning. Through further experiments, we find that reinforcement learning (RL) is a promising approach for enhancing the vision reasoning capabilities of MLLMs. To promote further research and innovation, we open-source the evaluation code, training scripts, and datasets associated with this work.

References

- [1] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [2] W. U. Ahmad, S. Narenthiran, S. Majumdar, A. Ficek, S. Jain, J. Huang, V. Noroozi, and B. Ginsburg. Opencodereasoning: Advancing data distillation for competitive coding. *arXiv preprint arXiv:2504.01943*, 2025.
- [3] A. Ahmadian, C. Cremer, M. Gallé, M. Fadaee, J. Kreutzer, O. Pietquin, A. stün, and S. Hooker. Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms, 2024.
- [4] S. N. Akter, S. Lee, Y. Chang, Y. Bisk, and E. Nyberg. Visreas: Complex visual reasoning with unanswerable questions, 2024.
- [5] J.-B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds, et al. Flamingo: a visual language model for few-shot learning. *dvances in neural information processing systems*, 35:23716–23736, 2022.
- [6] H. Bai, Y. Zhou, L. E. Li, S. Levine, and A. Kumar. Digi-q: Learning q-value functions for training device-control agents. *arXiv preprint arXiv:2502.15760*, 2025.
- [7] J. Bai, S. Bai, S. Yang, S. Wang, S. Tan, P. Wang, J. Lin, C. Zhou, and J. Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.
- [8] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang, H. Zhong, Y. Zhu, M. Yang, Z. Li, J. Wan, P. Wang, W. Ding, Z. Fu, Y. Xu, J. Ye, X. Zhang, T. Xie, Z. Cheng, H. Zhang, Z. Yang, H. Xu, and J. Lin. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- [9] G. Chen, M. Wang, Y. Yang, K. Yu, L. Yuan, and Y. Yue. Pointgpt: Auto-regressively generative pre-training from point clouds. *dvances in Neural Information Processing Systems*, 36:29667–29679, 2023.
- [10] L. Chen, J. Li, X. Dong, P. Zhang, C. He, J. Wang, F. Zhao, and D. Lin. Sharegpt4v: Improving large multi-modal models with better captions. *arXiv preprint arXiv:2311.12793*, 2023.
- [11] L. Chen, L. Li, H. Zhao, Y. Song, and Vinci. R1-v: Reinforcing super generalization ability in vision-language models with less than \$3. <https://github.com/Deep-gent/R1-V>, 2025. Accessed: 2025-02-02.
- [12] S. Chen, H. Li, Q. Wang, Z. Zhao, M. Sun, X. Zhu, and J. Liu. Vast: A vision-audio-subtitle-text omni-modality foundation model and dataset. *dvances in Neural Information Processing Systems*, 36:72842–72866, 2023.
- [13] X. Chen, H. Fang, T.-Y. Lin, R. Vedantam, S. Gupta, P. Dollar, and C. L. Zitnick. Microsoft coco captions: Data collection and evaluation server, 2015.
- [14] Z. Chen, W. Wang, Y. Cao, Y. Liu, Z. Gao, E. Cui, J. Zhu, S. Ye, H. Tian, Z. Liu, et al. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv preprint arXiv:2412.05271*, 2024.
- [15] Z. Chen, W. Wang, H. Tian, S. Ye, Z. Gao, E. Cui, W. Tong, K. Hu, J. Luo, Z. Ma, et al. How far are we to gpt-4v? closing the gap to commercial multimodal models with open-source suites. *arXiv preprint arXiv:2404.16821*, 2024.
- [16] Z. Chen, J. Wu, W. Wang, W. Su, G. Chen, S. Xing, M. Zhong, Q. Zhang, X. Zhu, L. Lu, et al. Intervl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24185–24198, 2024.
- [17] S. Cheng, Z. Guo, J. Wu, K. Fang, P. Li, H. Liu, and Y. Liu. Egothink: Evaluating first-person perspective thinking capability of vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14291–14302, 2024.
- [18] DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning, 2025.

- [19] A. Défossez, L. Mazaré, M. Orsini, A. Royer, P. Pérez, H. Jégou, E. Grave, and N. Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.
- [20] H. Dong, J. Li, B. Wu, J. Wang, Y. Zhang, and H. Guo. Benchmarking and improving detail image caption. *arXiv preprint arXiv:2405.19092*, 2024.
- [21] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- [22] Q. Fang, S. Guo, Y. Zhou, Z. Ma, S. Zhang, and Y. Feng. Llama-omni: Seamless speech interaction with large language models. *arXiv preprint arXiv:2409.06666*, 2024.
- [23] J. Feng, R. Xu, J. Hao, H. Sharma, Y. Shen, D. Zhao, and W. Chen. Language models can be logical solvers. *arXiv preprint arXiv:2311.06158*, 2023.
- [24] Z. Gao, Z. Chen, E. Cui, Y. Ren, W. Wang, J. Zhu, H. Tian, S. Ye, J. He, X. Zhu, et al. Mini-internvl: a flexible-transfer pocket multi-modal model with 5% parameters and 90% performance. *Visual Intelligence*, 2(1):1–17, 2024.
- [25] T. GLM, A. Zeng, B. Xu, B. Wang, C. Zhang, D. Yin, D. Rojas, G. Feng, H. Zhao, H. Lai, H. Yu, H. Wang, J. Sun, J. Zhang, J. Cheng, J. Gui, J. Tang, J. Zhang, J. Li, L. Zhao, L. Wu, L. Zhong, M. Liu, M. Huang, P. Zhang, Q. Zheng, R. Lu, S. Duan, S. Zhang, S. Cao, S. Yang, W. L. Tam, W. Zhao, X. Liu, X. Xia, X. Zhang, X. Gu, X. Lv, X. Liu, X. Liu, X. Yang, X. Song, X. Zhang, Y. An, Y. Xu, Y. Niu, Y. Yang, Y. Li, Y. Bai, Y. Dong, Z. Qi, Z. Wang, Z. Yang, Z. Du, Z. Hou, and Z. Wang. Chatglm: A family of large language models from glm-130b to glm-4 all tools, 2024.
- [26] B. Goertzel and C. Pennachin. *rtificial general intelligence*, volume 2. Springer, 2007.
- [27] Z. Guo, R. Zhang, X. Zhu, Y. Tang, X. Ma, J. Han, K. Chen, P. Gao, X. Li, H. Li, et al. Point-bind & point-llm: Aligning point cloud with multi-modality for 3d understanding, generation, and instruction following. *arXiv preprint arXiv:2309.00615*, 2023.
- [28] H. Gupta, S. Verma, U. Ananteswaran, K. Scaria, M. Parmar, S. Mishra, and C. Baral. Polymath: A challenging multi-modal mathematical reasoning benchmark, 2024.
- [29] X. Han, Q. You, Y. Liu, W. Chen, H. Zheng, K. Mrini, X. Lin, Y. Wang, B. Zhai, J. Yuan, et al. Infimm-eval: Complex open-ended reasoning evaluation for multi-modal large language models. *arXiv preprint arXiv:2311.11567*, 2023.
- [30] Y. Hao, J. Gu, H. W. Wang, L. Li, Z. Yang, L. Wang, and Y. Cheng. Can mllms reason in multimodality? emma: An enhanced multimodal reasoning benchmark. *arXiv preprint arXiv:2501.05444*, 2025.
- [31] C. He, R. Luo, Y. Bai, S. Hu, Z. L. Thai, J. Shen, J. Hu, X. Han, Y. Huang, Y. Zhang, et al. Olympiadbench: A challenging benchmark for promoting agi with olympiad-level bilingual multimodal scientific problems. *arXiv preprint arXiv:2402.14008*, 2024.
- [32] D. Huang, Q. Bu, Y. Qing, and H. Cui. Codecot: Tackling code syntax errors in cot reasoning for code generation. *arXiv preprint arXiv:2308.08784*, 2023.
- [33] A. Hurst, A. Lerer, A. P. Goucher, A. Perelman, A. Ramesh, A. Clark, A. Ostrow, A. Welihinda, A. Hayes, A. Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- [34] A. Jaech, A. Kalai, A. Lerer, A. Richardson, A. El-Kishky, A. Low, A. Helyar, A. Madry, A. Beutel, A. Carney, et al. Openai o1 system card. *arXiv preprint arXiv:2412.16720*, 2024.
- [35] X. Jiang, Y. Dong, L. Wang, Z. Fang, Q. Shang, G. Li, Z. Jin, and W. Jiao. Self-planning code generation with large language models. *CM Transactions on Software Engineering and Methodology*, 33(7):1–30, 2024.
- [36] W. Kay, J. Carreira, K. Simonyan, B. Zhang, C. Hillier, S. Vijayanarasimhan, F. Viola, T. Green, T. Back, P. Natsev, M. Suleyman, and A. Zisserman. The kinetics human action video dataset, 2017.
- [37] L. Ke, W. Pei, R. Li, X. Shen, and Y.-W. Tai. Reflective decoding network for image captioning. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 8888–8897, 2019.

- [38] X. Lai, Z. Tian, Y. Chen, Y. Li, Y. Yuan, S. Liu, and J. Jia. Lisa: Reasoning segmentation via large language model, 2024.
- [39] B. Li, Y. Zhang, D. Guo, R. Zhang, F. Li, H. Zhang, K. Zhang, P. Zhang, Y. Li, Z. Liu, and C. Li. Llava-onevision: Easy visual task transfer, 2024.
- [40] J. Li, D. Li, S. Savarese, and S. Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR, 2023.
- [41] J. Li, D. Li, C. Xiong, and S. Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*, pages 12888–12900. PMLR, 2022.
- [42] J. Li, G. Li, Y. Li, and Z. Jin. Structured chain-of-thought prompting for code generation. *CM Transactions on Software Engineering and Methodology*, 34(2):1–23, 2025.
- [43] J. Li, W. Lu, H. Fei, M. Luo, M. Dai, M. Xia, Y. Jin, Z. Gan, D. Qi, C. Fu, Y. Tai, W. Yang, Y. Wang, and C. Wang. A survey on benchmarks of multimodal large language models, 2024.
- [44] T.-Y. Lin, M. Maire, S. Belongie, L. Bourdev, R. Girshick, J. Hays, P. Perona, D. Ramanan, C. L. Zitnick, and P. Dollár. Microsoft coco: Common objects in context, 2015.
- [45] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual instruction tuning. *dvances in neural information processing systems*, 36:34892–34916, 2023.
- [46] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual instruction tuning, 2023.
- [47] H. Liu, Z. Teng, L. Cui, C. Zhang, Q. Zhou, and Y. Zhang. Logicot: Logical chain-of-thought instruction-tuning. *arXiv preprint arXiv:2305.12147*, 2023.
- [48] X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang, et al. Agentbench: Evaluating llms as agents. *arXiv preprint arXiv:2308.03688*, 2023.
- [49] Y. Liu, Z. Li, M. Huang, B. Yang, W. Yu, C. Li, X.-C. Yin, C.-L. Liu, L. Jin, and X. Bai. Ocrbench: on the hidden mystery of ocr in large multimodal models. *Science China Information Sciences*, 67(12):220102, 2024.
- [50] Z. Liu, Z. Sun, Y. Zang, X. Dong, Y. Cao, H. Duan, D. Lin, and J. Wang. Visual-rft: Visual reinforcement fine-tuning. *arXiv preprint arXiv:2503.01785*, 2025.
- [51] Z. Liu, Y. Zhang, F. Liu, C. Zhang, Y. Sun, and J. Wang. Othink-mr1: Stimulating multimodal generalized reasoning capabilities through dynamic reinforcement learning. *arXiv preprint arXiv:2503.16081*, 2025.
- [52] P. Lu, H. Bansal, T. Xia, J. Liu, C. Li, H. Hajishirzi, H. Cheng, K.-W. Chang, M. Galley, and J. Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023.
- [53] S. Lu, Y. Li, Q.-G. Chen, Z. Xu, W. Luo, K. Zhang, and H.-J. Ye. Ovis: Structural embedding alignment for multimodal large language model. *arXiv:2405.20797*, 2024.
- [54] K. Mangalam, R. Akshulakov, and J. Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. *dvances in Neural Information Processing Systems*, 36:46212–46244, 2023.
- [55] J. Mao, J. Huang, A. Toshev, O. Camburu, A. L. Yuille, and K. Murphy. Generation and comprehension of unambiguous object descriptions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 11–20, 2016.
- [56] A. Masry, D. X. Long, J. Q. Tan, S. Joty, and E. Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. *arXiv preprint arXiv:2203.10244*, 2022.
- [57] M. Mathew, D. Karatzas, and C. Jawahar. Docvqa: A dataset for vqa on document images. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 2200–2209, 2021.
- [58] F. Meng, L. Du, Z. Liu, Z. Zhou, Q. Lu, D. Fu, B. Shi, W. Wang, J. He, K. Zhang, et al. Mm-eureka: Exploring visual aha moment with rule-based large-scale reinforcement learning. *arXiv preprint arXiv:2503.07365*, 2025.

- [59] T. Nguyen, S. Y. Gadre, G. Ilharco, S. Oh, and L. Schmidt. Improving multimodal datasets with image captioning. *Advances in Neural Information Processing Systems*, 36:22047–22069, 2023.
- [60] R. Niu, J. Li, S. Wang, Y. Fu, X. Hu, X. Leng, H. Kong, Y. Chang, and Q. Wang. Screenagent: A vision language model-driven computer control agent. *arXiv preprint arXiv:2402.07945*, 2024.
- [61] Y. Peng, G. Zhang, M. Zhang, Z. You, J. Liu, Q. Zhu, K. Yang, X. Xu, X. Geng, and X. Yang. Lmm-r1: Empowering 3b llms with strong reasoning abilities through two-stage rule-based rl. *arXiv preprint arXiv:2503.07536*, 2025.
- [62] R. Qiao, Q. Tan, G. Dong, M. Wu, C. Sun, X. Song, Z. GongQue, S. Lei, Z. Wei, M. Zhang, et al. We-math: Does your large multimodal model achieve human-like mathematical reasoning? *arXiv preprint arXiv:2407.01284*, 2024.
- [63] H. Shen, Z. Zhang, K. Zhao, Q. Zhang, R. Xu, and T. Zhao. Vlm-r1: A stable and generalizable r1-style large vision-language model. <https://github.com/om-ai-lab/VLM-R1>, 2025. Accessed: 2025-02-15.
- [64] G. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- [65] K. Team, A. Du, B. Gao, B. Xing, C. Jiang, C. Chen, C. Li, C. Xiao, C. Du, C. Liao, et al. Kimi k1. 5: Scaling reinforcement learning with llms. *arXiv preprint arXiv:2501.12599*, 2025.
- [66] Q. Team. Qvq: To see the world with wisdom, December 2024.
- [67] Q. Team. Qwq-32b: Embracing the power of reinforcement learning, March 2025.
- [68] Y. Wan, W. Wang, Y. Yang, Y. Yuan, J.-t. Huang, P. He, W. Jiao, and M. R. Lyu. Logicasker: Evaluating and improving the logical reasoning ability of large language models. *arXiv preprint arXiv:2401.00757*, 2024.
- [69] K. Wang, J. Pan, W. Shi, Z. Lu, H. Ren, A. Zhou, M. Zhan, and H. Li. Measuring multimodal mathematical reasoning with math-vision dataset. *Advances in Neural Information Processing Systems*, 37:95095–95169, 2024.
- [70] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.
- [71] S. Wang, D. Kim, A. Taalimi, C. Sun, and W. Kuo. Learning visual grounding from generative vision and language model. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 8057–8067. IEEE, 2025.
- [72] W. Wang, Z. Gao, L. Chen, Z. Chen, J. Zhu, X. Zhao, Y. Liu, Y. Cao, S. Ye, X. Zhu, et al. Visualprm: An effective process reward model for multimodal reasoning. *arXiv preprint arXiv:2503.10291*, 2025.
- [73] X. Wang and P. Peng. Open-r1-video. <https://github.com/Wang-Xiaodong1899/Open-R1-Video>, 2025.
- [74] Y. Wang, W. Chen, X. Han, X. Lin, H. Zhao, Y. Liu, B. Zhai, J. Yuan, Q. You, and H. Yang. Exploring the reasoning abilities of multimodal large language models (mllms): A comprehensive survey on emerging trends in multimodal reasoning, 2024.
- [75] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.
- [76] Y. Xiao, E. Sun, T. Liu, and W. Wang. Logicvista: Multimodal llm logical reasoning benchmark in visual contexts. *arXiv preprint arXiv:2407.04973*, 2024.
- [77] Z. Xie and C. Wu. Mini-omni: Language models can hear, talk while thinking in streaming, 2024. URL <https://arxiv.org/abs/2408.16725>, 2024.
- [78] D. Xu, Z. Zhao, J. Xiao, F. Wu, H. Zhang, X. He, and Y. Zhuang. Video question answering via gradually refined attention over appearance and motion. In *Proceedings of the 25th ACM international conference on Multimedia*, pages 1645–1653, 2017.

- [79] F. Xu, Z. Wu, Q. Sun, S. Ren, F. Yuan, S. Yuan, Q. Lin, Y. Qiao, and J. Liu. Symbol-llm: Towards foundational symbol-centric interface for large language models. *arXiv preprint arXiv:2311.09278*, 2023.
- [80] R. Xu, Z. Huang, T. Wang, Y. Chen, J. Pang, and D. Lin. Vlm-grounder: A vlm agent for zero-shot 3d visual grounding. *arXiv preprint arXiv:2410.13860*, 2024.
- [81] Y. Xu, X. Liu, X. Liu, Z. Hou, Y. Li, X. Zhang, Z. Wang, A. Zeng, Z. Du, W. Zhao, et al. Chatglm-math: Improving math problem-solving in large language models with a self-critique pipeline. *arXiv preprint arXiv:2404.02893*, 2024.
- [82] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [83] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, H. Lin, J. Yang, J. Tu, J. Zhang, J. Yang, J. Yang, J. Zhou, J. Lin, K. Dang, K. Lu, K. Bao, K. Yang, L. Yu, M. Li, M. Xue, P. Zhang, Q. Zhu, R. Men, R. Lin, T. Li, T. Xia, X. Ren, X. Ren, Y. Fan, Y. Su, Y. Zhang, Y. Wan, Y. Liu, Z. Cui, Z. Zhang, and Z. Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [84] Y. Yang, X. He, H. Pan, X. Jiang, Y. Deng, X. Yang, H. Lu, D. Yin, F. Rao, M. Zhu, B. Zhang, and W. Chen. R1-onevision: Advancing generalized multimodal reasoning through cross-modal formalization. *arXiv preprint arXiv:2503.10615*, 2025.
- [85] Y. Yao, T. Yu, A. Zhang, C. Wang, J. Cui, H. Zhu, T. Cai, H. Li, W. Zhao, Z. He, et al. Minicpm-v: A gpt-4v level mllm on your phone. *arXiv preprint arXiv:2408.01800*, 2024.
- [86] J. Ye, G. Li, S. Gao, C. Huang, Y. Wu, S. Li, X. Fan, S. Dou, Q. Zhang, T. Gui, et al. Tooleyes: fine-grained evaluation for tool learning capabilities of large language models in real-world scenarios. *arXiv preprint arXiv:2401.00741*, 2024.
- [87] J. Ye, H. Xu, H. Liu, A. Hu, M. Yan, Q. Qian, J. Zhang, F. Huang, and J. Zhou. mplug-owl3: Towards long image-sequence understanding in multi-modal large language models, 2024.
- [88] L. Yu, P. Poirson, S. Yang, A. C. Berg, and T. L. Berg. Modeling context in referring expressions. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part II 14*, pages 69–85. Springer, 2016.
- [89] X. Yue, Y. Ni, K. Zhang, T. Zheng, R. Liu, G. Zhang, S. Stevens, D. Jiang, W. Ren, Y. Sun, C. Wei, B. Yu, R. Yuan, R. Sun, M. Yin, B. Zheng, Z. Yang, Y. Liu, W. Huang, H. Sun, Y. Su, and W. Chen. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of CVPR*, 2024.
- [90] C. Zhang, F. Gao, B. Jia, Y. Zhu, and S.-C. Zhu. Raven: A dataset for relational and analogical visual reasoning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [91] R. Zhang, D. Jiang, Y. Zhang, H. Lin, Z. Guo, P. Qiu, A. Zhou, P. Lu, K.-W. Chang, Y. Qiao, et al. Mathverse: Does your multi-modal llm truly see the diagrams in visual math problems? In *European Conference on Computer Vision*, pages 169–186. Springer, 2024.
- [92] Q. Zhao, S. Wang, C. Zhang, C. Fu, M. Q. Do, N. Agarwal, K. Lee, and C. Sun. Antgpt: Can large language models help long-term action anticipation from videos? *arXiv preprint arXiv:2307.16368*, 2023.
- [93] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592*, 2023.
- [94] J. Zhu, W. Wang, Z. Chen, Z. Liu, S. Ye, L. Gu, Y. Duan, H. Tian, W. Su, J. Shao, Z. Gao, E. Cui, Y. Cao, Y. Liu, X. Wei, H. Zhang, H. Wang, W. Xu, H. Li, J. Wang, D. Chen, S. Li, Y. He, T. Jiang, J. Luo, Y. Wang, C. He, B. Shi, X. Zhang, W. Shao, J. He, Y. Xiong, W. Qu, P. Sun, P. Jiao, H. Lv, L. Wu, K. Zhang, H. Deng, J. Ge, K. Chen, L. Wang, M. Dou, L. Lu, X. Zhu, T. Lu, D. Lin, Y. Qiao, J. Dai, and W. Wang. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models, 2025.

Overview of the appendix

In the appendix, we provide additional details and supplementary information to further elaborate on sections mentioned above. In Section B, we analyze the statistical features of the dataset, meanwhile providing examples of questions ranging from different categories. Section C contains experiments details, including the evaluation of LLMs, the evaluation of hint prompts and RL experiments. Some examples of model outputs are also illustrated.

B Benchmark analysis

B.1 Statistical analysis

As shown in Figure 10, the text length of questions in VisuLogic is mostly concentrated around 40 tokens (calculated by Llama-3.1’s and InternVL2.5’s tokenizer). We also analyze the distribution of image sizes, as shown in Figure 9. The image widths range from 200 to 700 pixels, with an average of 592.3 pixels, while the heights range from 90 to 825 pixels, with an average of 327.9 pixels.

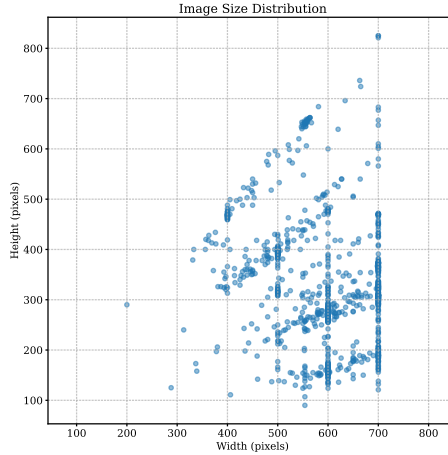


Figure 9: **Image size distribution.** The size of images is limited to within the same order of magnitude.

B.2 More Examples of VisuLogic

To provide a thoroughly presentation of our benchmark, we include more examples of questions from different categories in the Figure 11 and Figure 12.

C Evaluation & Experiment

C.1 Evaluation of LLMs

Caption generation for LLMs Evaluation. In our experiment, we employ large language models (LLMs) for comparative analysis. Specifically, when setting up the LLM-based experiment, we initially utilize GPT-4o to generate captions for images with the following prompt: *Please describe the fine-grained content of the image or figure based on this question, including scenes, objects, relationships, and any text present. Please note that you do not need to answer this question directly, just describe the information of this picture.* Additional examples of generated image captions are presented in Figure 14 and Figure 15.

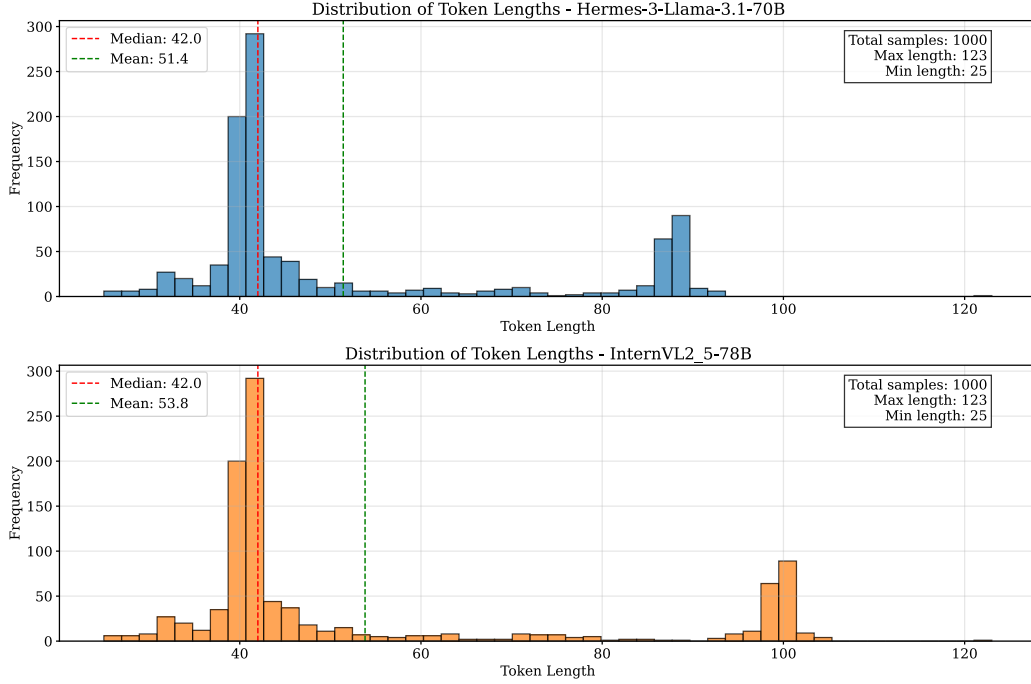


Figure 10: Distribution of text token length in VisuLogic.

More Examples of Captions. We provide additional image captions for six categories, as illustrated in Figures 14 and 15. Even SOTA MLLM (GPT-4o) encounters difficulties in accurately describing the details of images from VisuLogic.

C.2 More Solutions from Models

We provide more solutions generated from different LLMs/MLLMs on our benchmark, as shown in Figure 16, Figure 17 and Figure 18. For the majority of questions, almost all models fail to provide accurate solutions. Sometimes even when the final answer is correct, methodological wrong may persist.

C.3 Hint Prompts Evaluation Details

We first generate hint prompts with GPT-4o, combining reference solutions with question data as inputs. All outputs undergo manual validation to prevent solution leakage. More examples are shown in Figure 19. After that, we input the hint prompts along with the same CoT prompt in CoT experiments (“*Solve the complex visual logical reasoning problem through step-by-step reasoning. Think about the reasoning process first and answer the question following this format: `answer: \boxed{$LETTER}`.*”) to MLLMs.

C.4 RL Experiments

Comparative SFT Experiments. To verify the effectiveness of RL method, we arrange the comparative SFT experiments on the same dataset as RL experiments. The instruction consists of questions and Non-CoT prompts, and the responses are formatted direct answers.

RL Algorithm. We employ REINFORCE Leave-One-Out (RLOO) [3] in our reinforcement learning training phase. As a critic-model-free algorithm, rloo is at a low computational cost while maintaining more robustness to noise and KL constraints.

Quantitative Reasoning

From the four given options, select the most suitable one to fill in the question mark.

From the given four options, choose the most suitable one to fill in the question mark.

Divide the following figures into two categories, ensuring that each category shares common patterns.

A: ①②⑥, ③④⑤
B: ①④⑤, ②③⑥
C: ①③⑥, ②④⑤
D: ①③⑤, ②④⑥

From the given four options, select the most suitable one to fill in the question mark.

From the given four options, select the most suitable one to fill in the question mark.

From the given four options, select the most appropriate one to fill in the question mark.

Spatial Reasoning

From any angle, which one on the right is not a view of the three-dimensional shape given on the left?

Which of the following four options is the front elevation view of the figure in the question.

The image on the left shows the net of a cube. Which of the options on the right can be formed by folding it?

The diagram on the left shows the net of a cube's outer surface. Please identify the incorrect option.

Choose the option from the four given choices that cannot assemble into the 3D shape shown in the question.

The left shows the net of a cube's outer surface. Which option on the right can be folded into it?

Positional Reasoning

From the four given options, select the most suitable one to fill in the question mark.

From the four given options, choose the most suitable one to fill in the question mark to create a certain pattern.

From the four given options, choose the most appropriate one to fill in the question mark.

The option that best matches the given pattern is ().

From the given four options, choose the most suitable one to fill in the question mark.

From the given four options, choose the most suitable one to present a certain regularity.

Figure 11: More examples in VisuLogic of Quantitative Reasoning, Spatial Reasoning, Positional Reasoning.

Attribute Reasoning

Divide the six figures below into two categories, so that each category of figures has its own common patterns.

A: ①②③, ④⑤⑥
B: ①②⑤, ③④⑥
C: ①③⑤, ②④⑥
D: ①④⑥, ②③⑤

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

Divide the following figures into two categories, ensuring that each category shares common patterns.

A: ①②⑤, ③④⑥
B: ①③⑥, ②④⑤
C: ①②④, ③⑤⑥
D: ①③④, ②⑤⑥

Divide the six figures below into two categories, so that each category of figures has its own common patterns.

A: ①②③, ④⑤⑥
B: ①②⑥, ③④⑤
C: ①③④, ②⑤⑥
D: ①④⑤, ②③⑥

From the given four options, select the most suitable one to fill in the question mark.

A B C D

From the given four options, select the most appropriate one to fill in the question mark.

A B C D

Stylistic Reasoning

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

Select the most suitable option from the four given choices to fill in the question mark.

A B C D

Choose the most suitable option from the four given choices to fill in the question mark.

A B C D

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

From the four given options, choose the most appropriate one to fill in the question mark.

A B C D

Other

From the four given options, choose the most suitable one to fill in the question mark: A: 2 B: 6 C: 4 D: 3.

A B C D

From the given four options, choose the most appropriate one to fill in the question mark.

A B C D

Select the most suitable option from the four given choices to fill in the question mark.

A B C D

Choose the most suitable option from the four given choices to fill in the question mark.

A B C D

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

Figure 12: More examples in VisuLogic of Attribute Reasoning, Stylistic Reasoning, and Other.

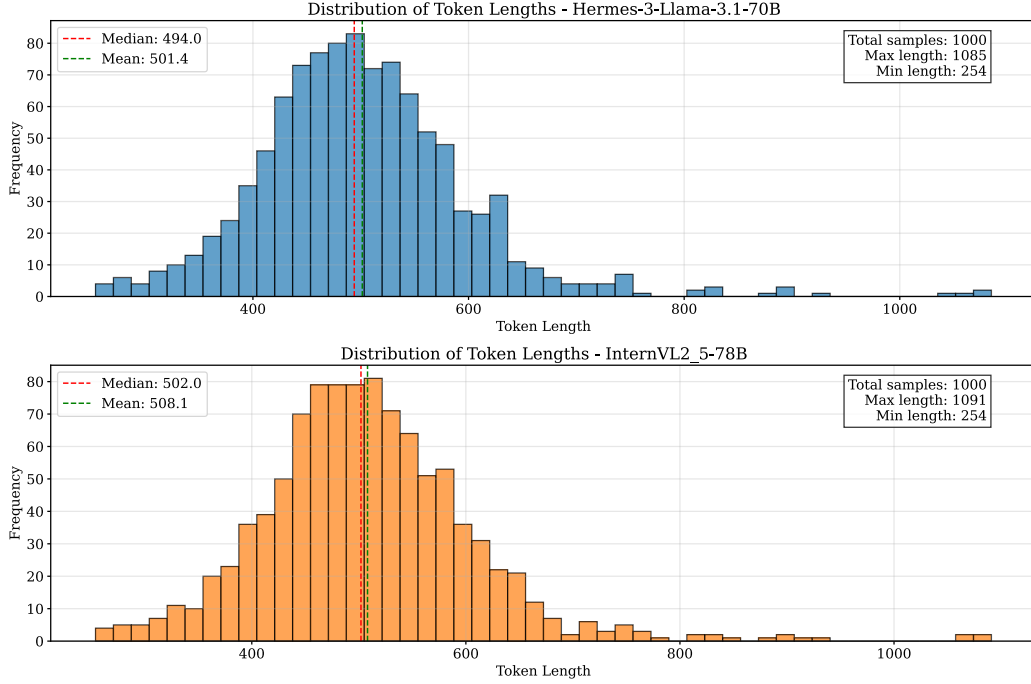


Figure 13: Distribution of tokens length in LLM evaluation settings, including image description.

Reward Modeling. Inspired by Deepseek-R1 [18], we design our rule-based reward system that mainly consists of two types of rewards:

1. **Format rewards:** To clarify model’s outputs, we design a format rule that forces model to put its thinking process between ‘<think>’ and ‘</think>’ tags and put its final answer between ‘<answer>’ and ‘</answer>’ tags. Regular expression is applied to judge whether outputs conform to the format rule.
2. **Accuracy rewards:** The accuracy reward is decided by the response’s correctness. The model should generate the response in right format, then the answer is extracted and judged whether it is matched to the correct option.

Hyperparameter settings. Our two RL models are trained with the hyperparameter configuration detailed in Table 5. And the hyperparameters used in SFT training stage are listed in Table 4.

Table 4: Hyperparameter Settings for SFT Training Stage.

	Qwen2.5-VL-7B-Instruct-SFT	InternVL2.5-38B-SFT
pretrain model	Qwen2.5-VL-7B-Instruct	InternVL2.5-38B
learning rate	0.5e-5	2e-5
batch size	64	128
optimizer	AdamW	AdamW
lr scheduler	cosine	cosine
image strategy	image_max_pixels=262144	max_dynamic_patch=6
warmup ratio	0.1	0.03
max epochs	1	1
bf16	True	True

Table 5: Hyperparameter Settings for RL Training Stage.

	Qwen2.5-VL-7B-Instruct-RL	InternVL2.5-38B-RL
pretrain model	Qwen2.5-VL-7B-Instruct	InternVL2.5-38B
RL Algorithm	rloo	rloo
train batch size	128	64
rollout batch_size	256	128
temperature	1	1
n samples per prompt	16	8
prompt max len	1024	4096
generate max len	3000	3000
bf16	True	True
actor learning rate	1e-6	1e-6
init kl coef	0	0

Other Details. The training environment consists of CentOS Linux release 7.6.1810 operating system with CUDA 12.1. For Qwen2.5-VL-7B-Instruct-RL, we train for 80 steps on 1 8 A800 GPUs and for InternVL2.5-38B-RL we train for 100 steps on 6 8 A800 GPUs.

C.5 RL models Evaluation Details

As mentioned above, we apply format rewards in RL experiments. Thus, to fully investigate the models’ latent reasoning abilities, we utilize implement training-aligned prompts during evaluation in VisuLogic, which is shown as follows: “*Solve the complex visual logical reasoning problem through step-by-step reasoning. Think about the reasoning process first and answer the question following this format: <think> THINKING </think><answer> NSWER </answer>*”.

C.6 Effectiveness of RL Experiments

Figures 20, 21, 22, 23, 24 and 25 demonstrate qualitative differences in model outputs between baseline and RL optimized models. It illustrates reinforcement learning (RL) training enables the model to perform fundamental-level analysis of reasoning tasks embedded in graphical representations.

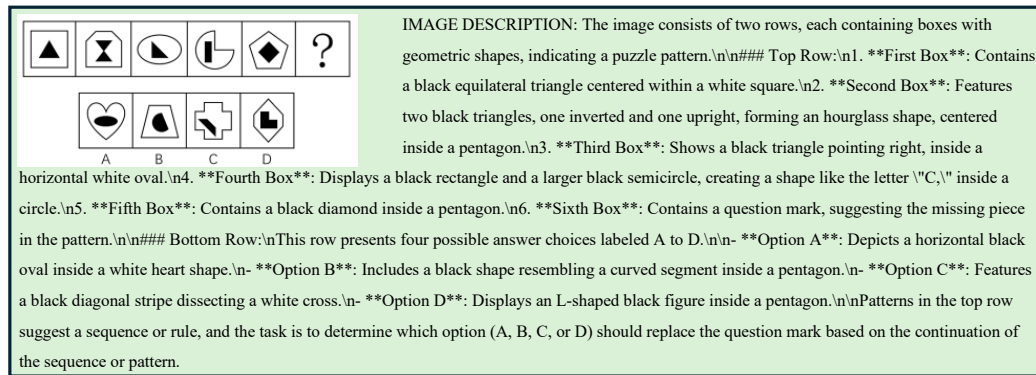
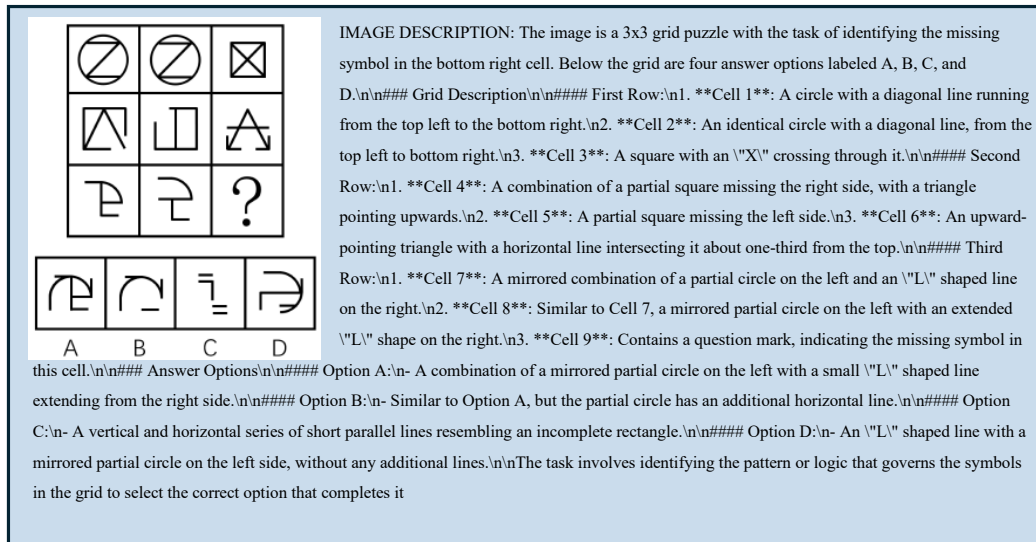
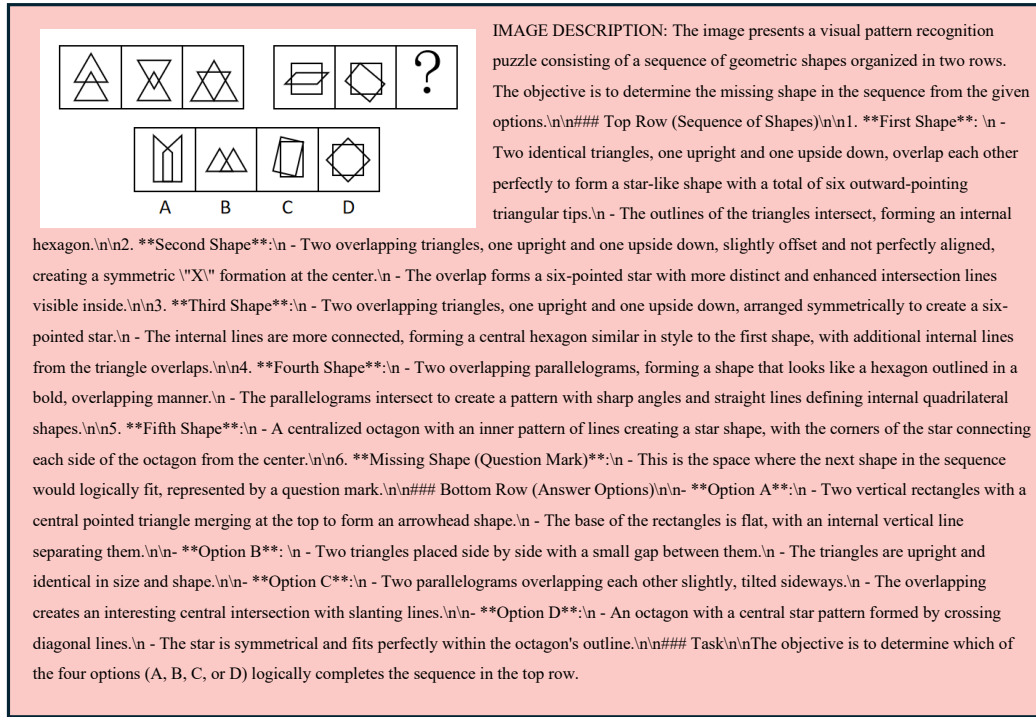


Figure 14: Part of image caption in LLM evaluation.

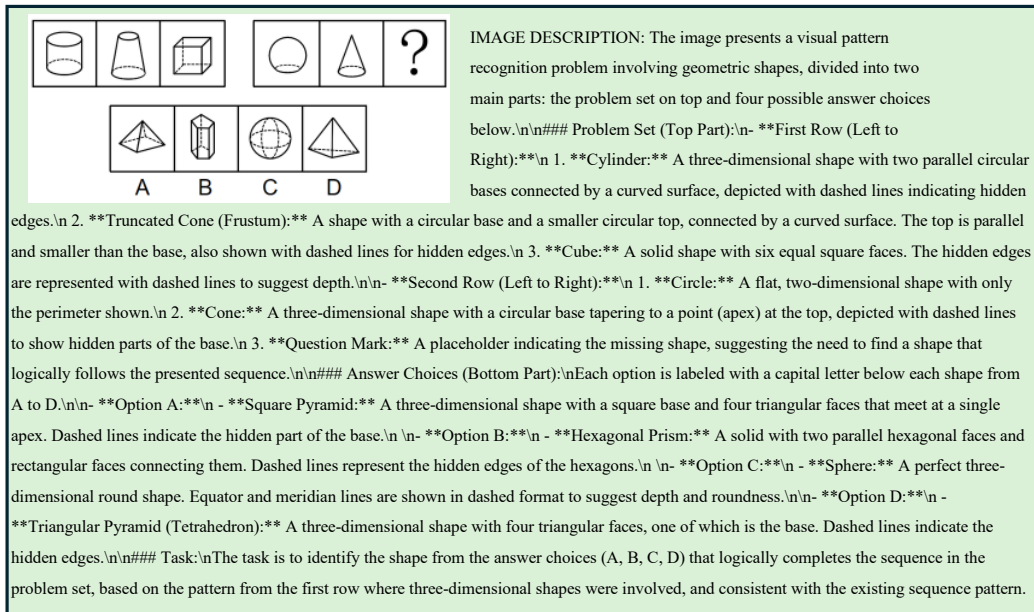
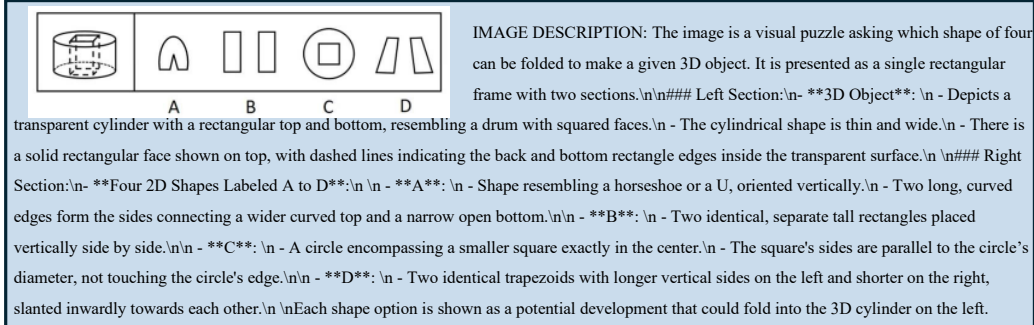
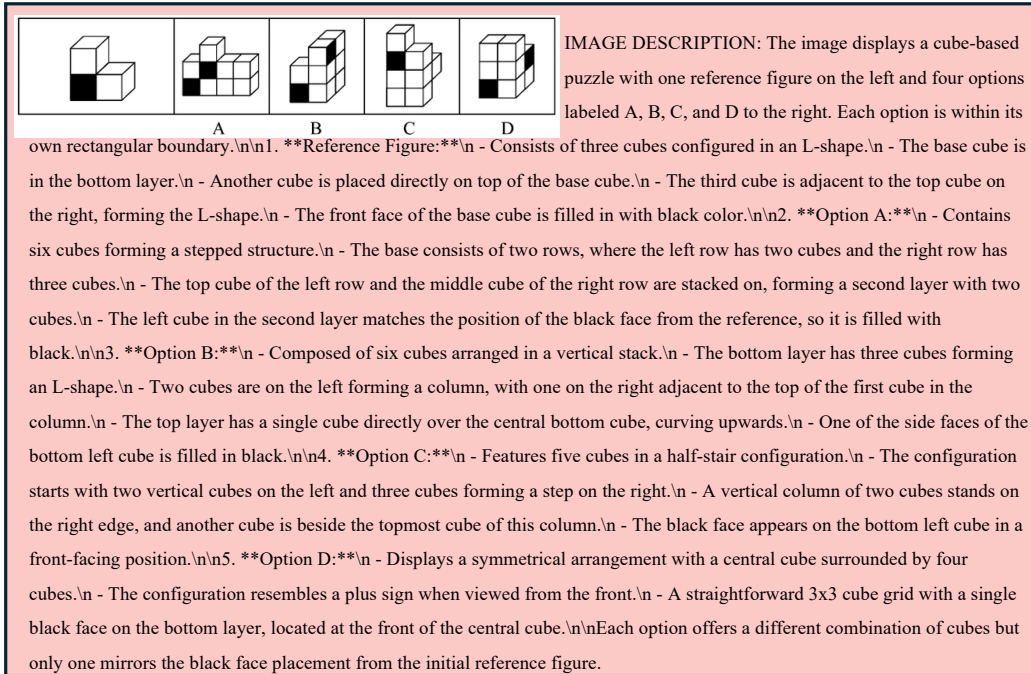


Figure 15: Part of image caption in LLM evaluation.

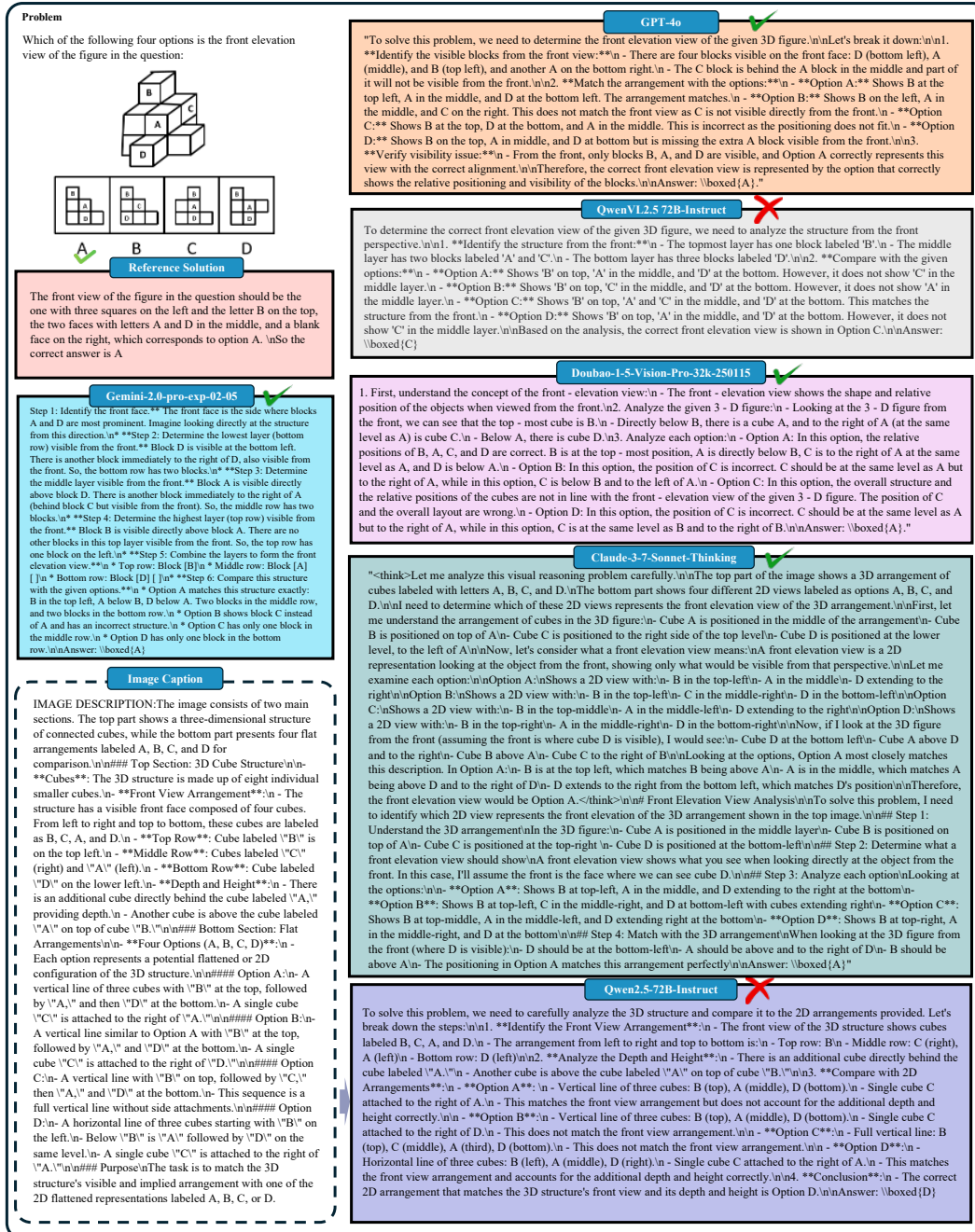


Figure 16: **Solution examples generated by different models.** Reference solution and outputs generated by GPT-4o, Qwen2.5VL-72B-Instruct, Gemini-2.0-pro-exp-02-05, Doubao-1.5-Vision-Pro-32K and Claude-3.7-sonnet-thinking. Additionally, the image caption and solution from LLMs (Qwen2.5-72B-Instruct) are also illustrated.

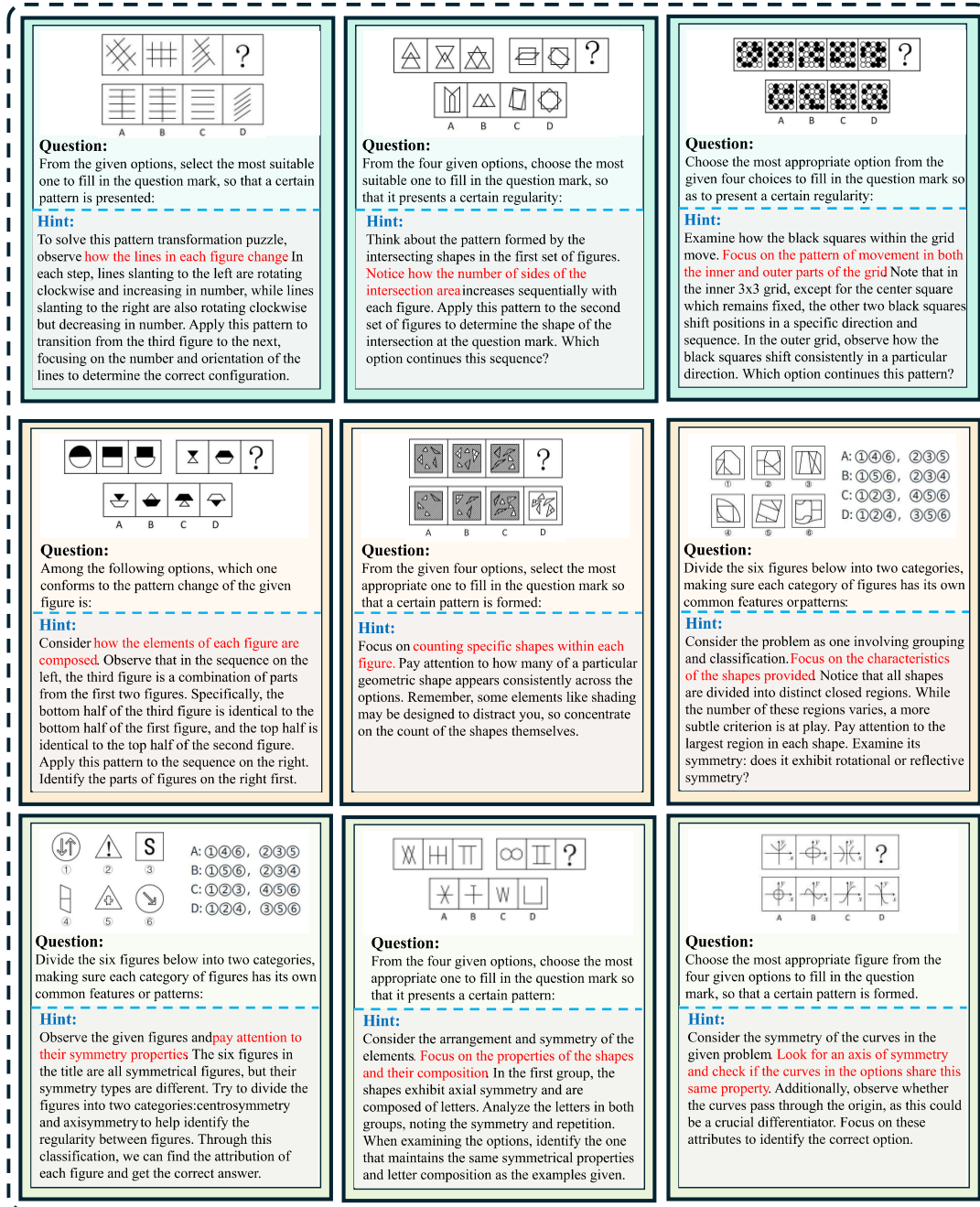


Figure 19: Examples of hint prompts. Hint prompts are provided to guide reasoning without revealing the final answer directly.

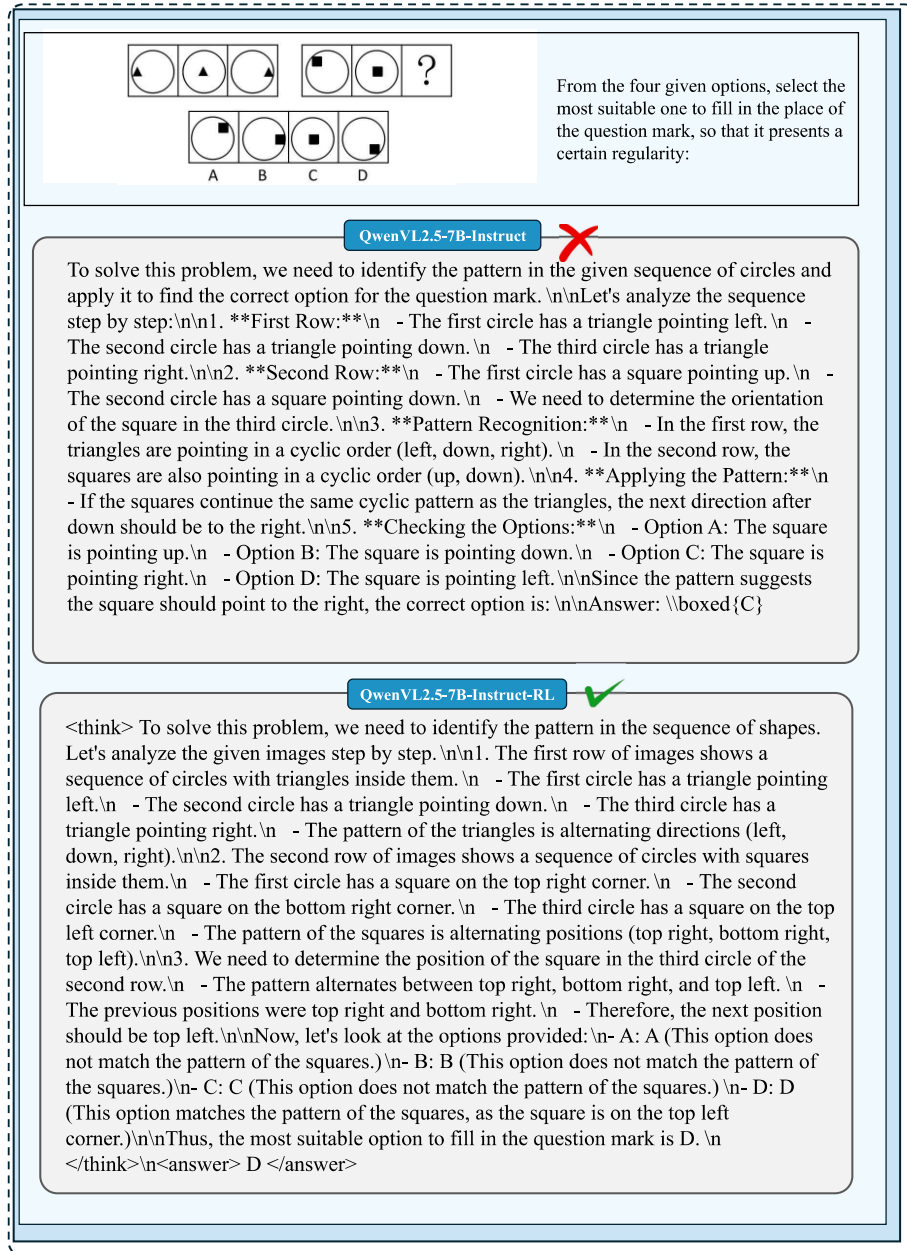


Figure 20: Comparison of model outputs before and after RL training stage for Qwen2.5-VL-7B.

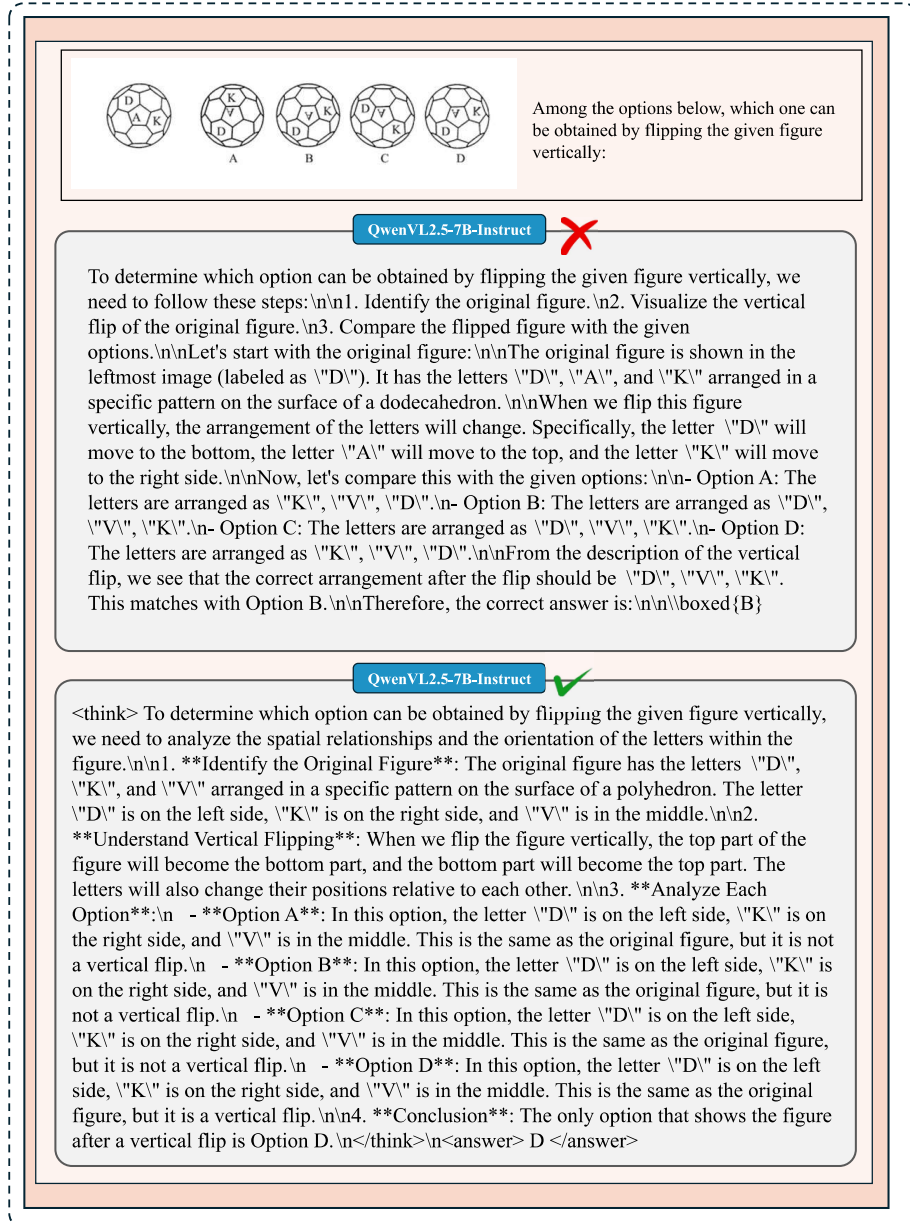


Figure 21: Comparison of model outputs before and after RL training stage for Qwen2.5-VL-7B.

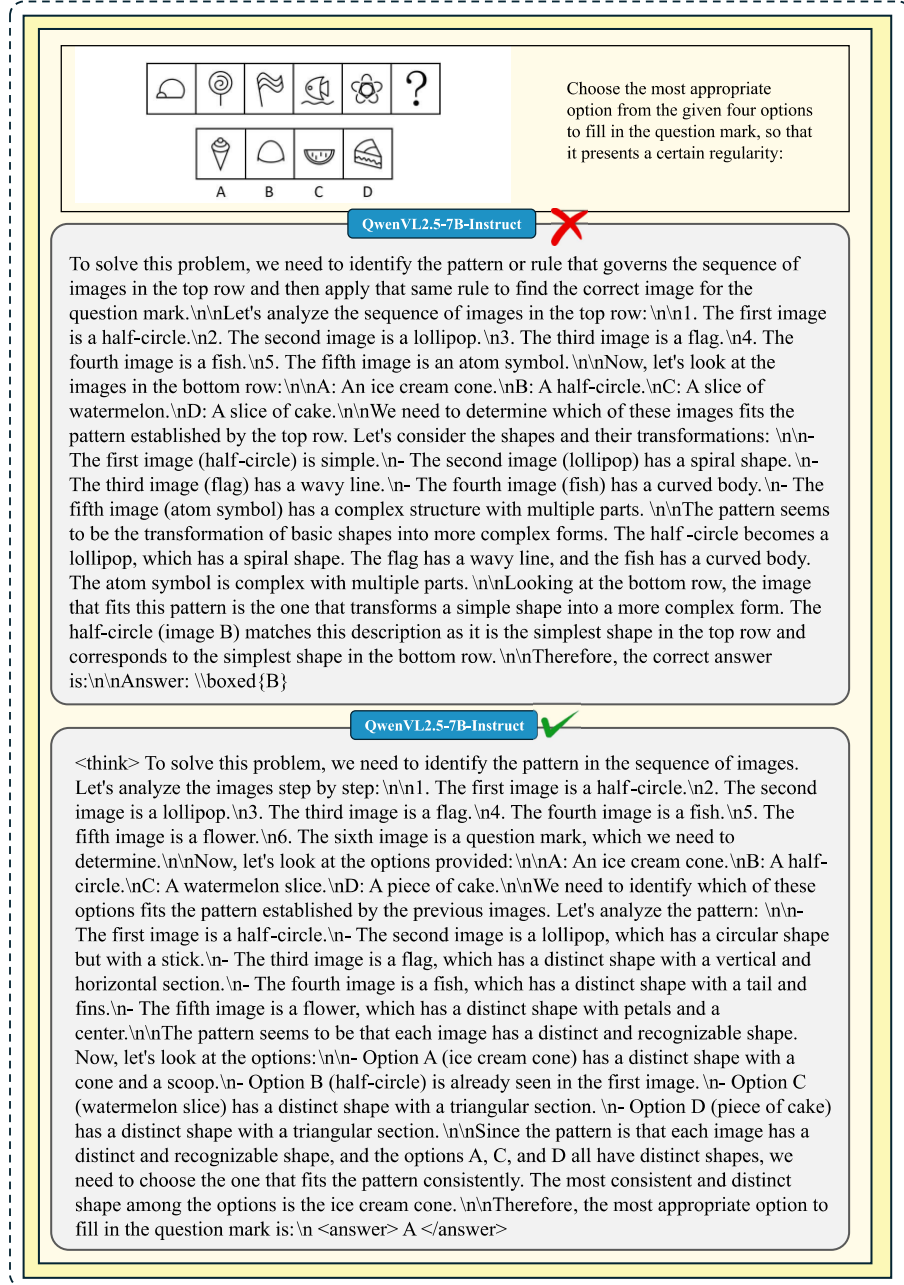


Figure 22: Comparison of model outputs before and after RL training stage for Qwen2.5-VL-7B.

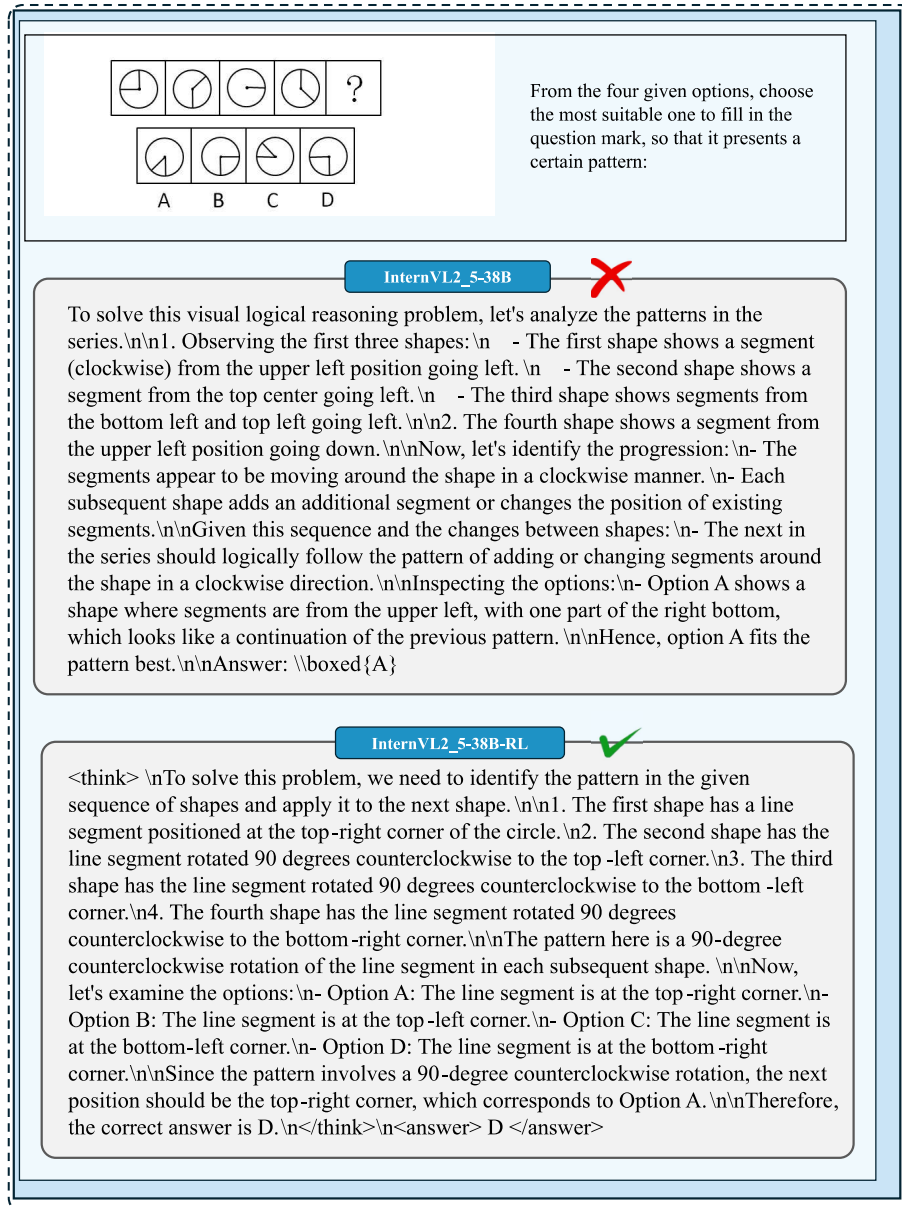


Figure 23: Comparison of model outputs before and after RL training stage for InternVL-2.5-38B.

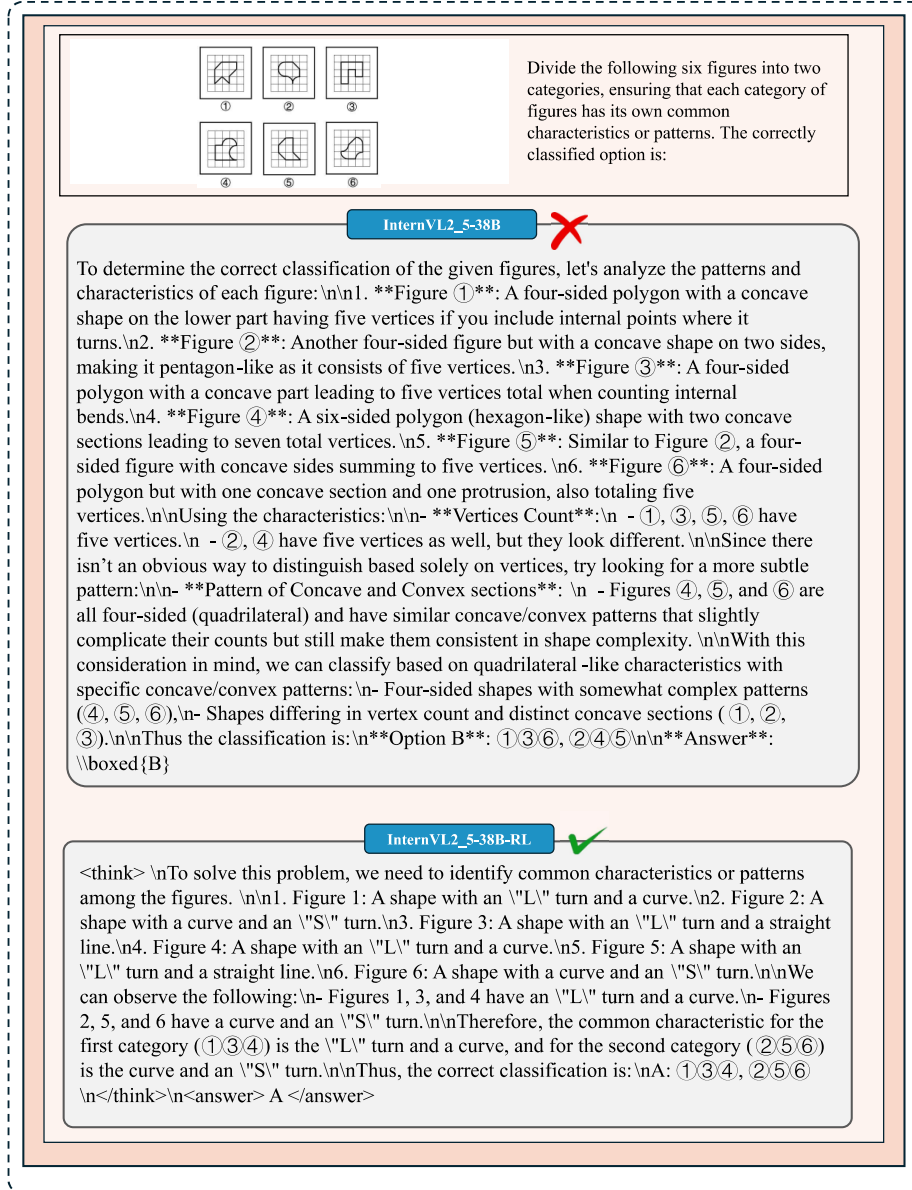


Figure 24: Comparison of model outputs before and after RL training stage for InternVL-2.5-38B.

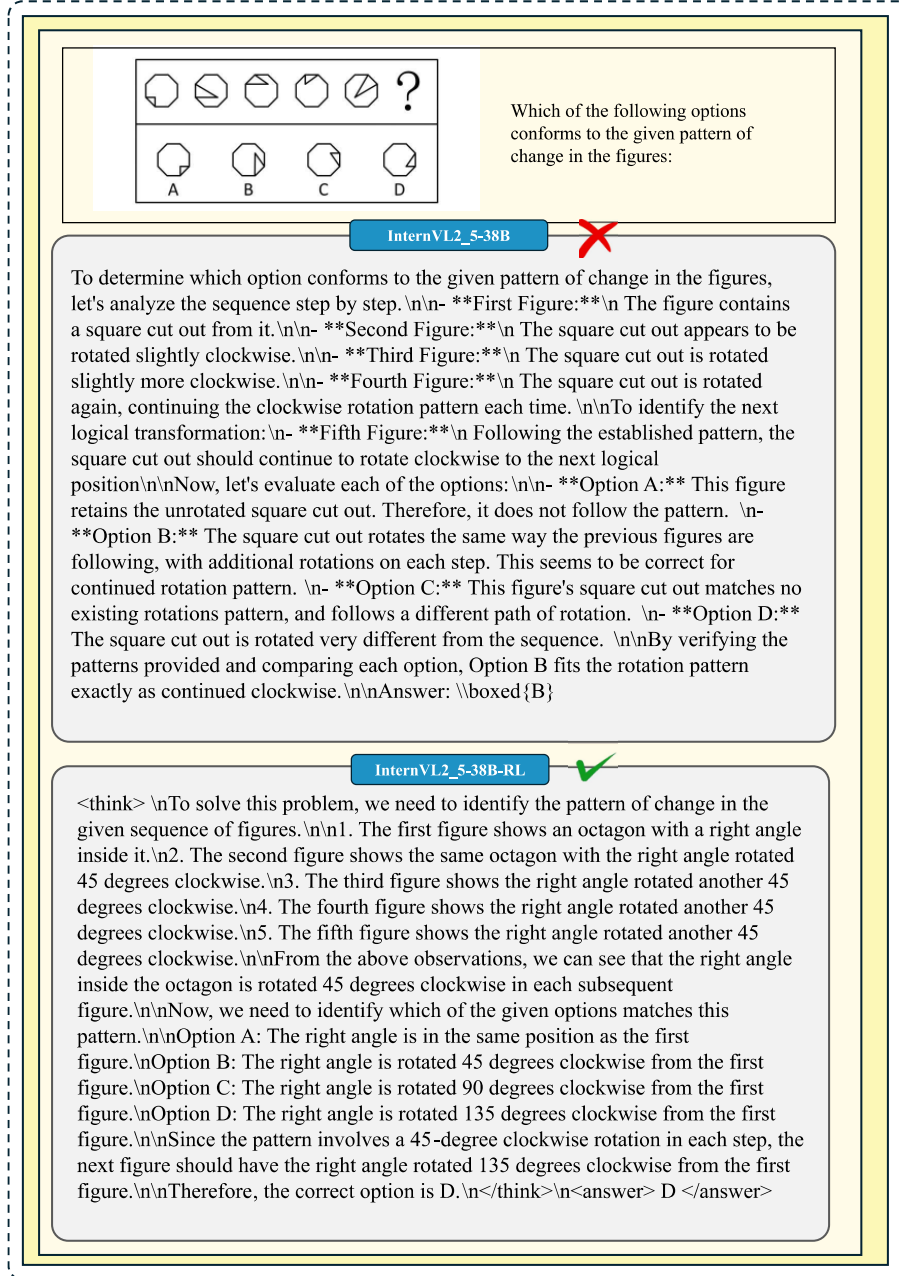


Figure 25: Comparison of model outputs before and after RL training stage for InternVL-2.5-38B.